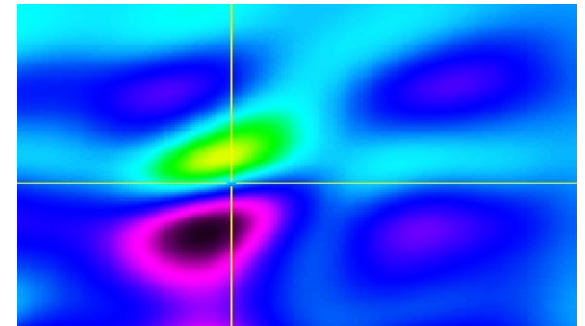
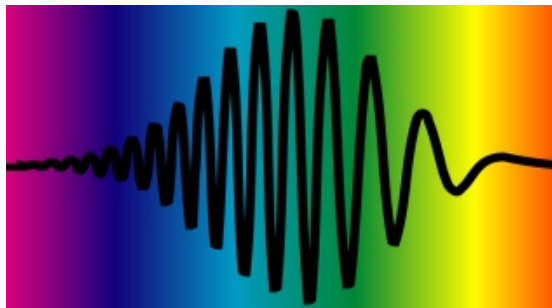


Coherent Spectroscopy of Quantum Dots : Experiments & Theory



Madhuri Mukhopadhyay

Works done in:

IACS, Kolkata

Lund University, Sweden

University of Arkansas, USA

Georgia Institute of Technology, USA

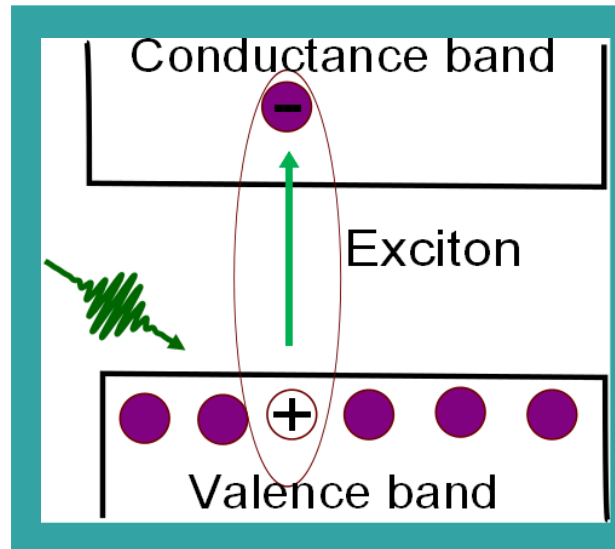
JNCASR, Bangalore, India

IISC, Bangalore, India

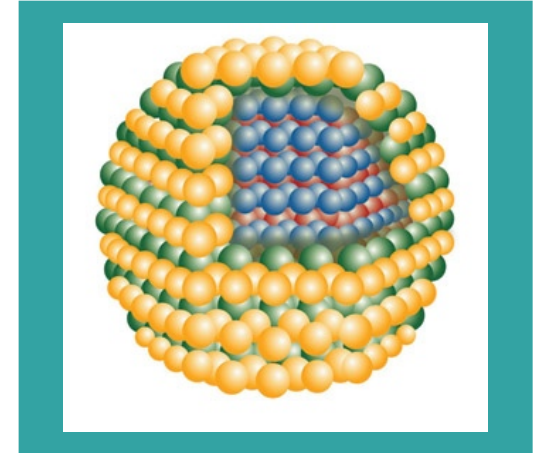
Introduction



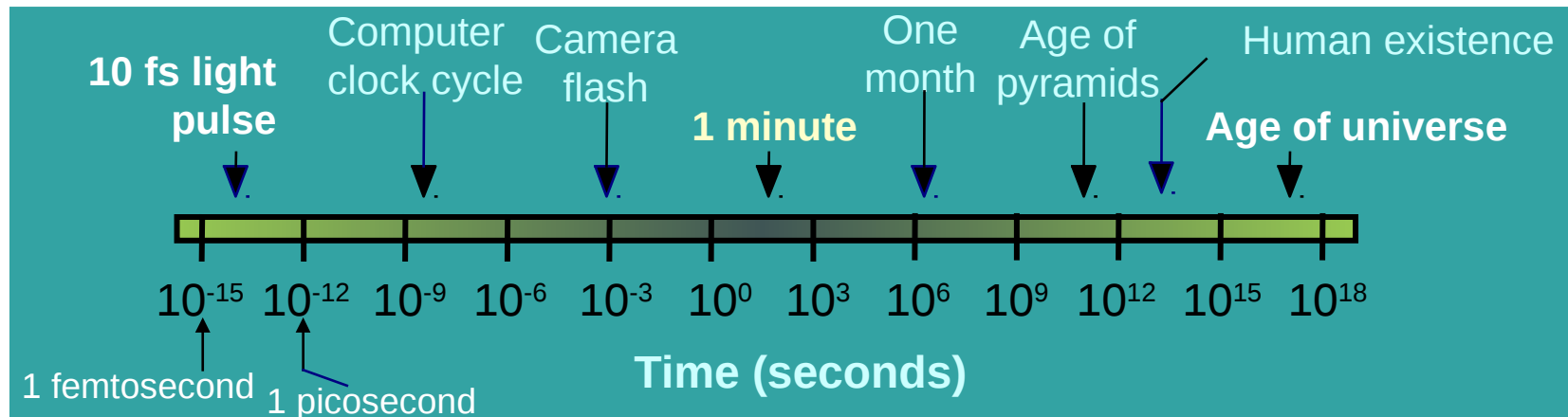
Solar to chemical
energy conversion



Excitation



Quantum dot



Plan of the Presentation

A. Techniques

1. Ultrafast: how fast & how do we measure
2. Coherent spectroscopy
3. Theoretical tools

B. Systems & Observations

1. Experimental studies
2. Theoretical studies

C. Conclusion

D. Future outlook

Techniques

Ultrafast coherent Spectroscopy

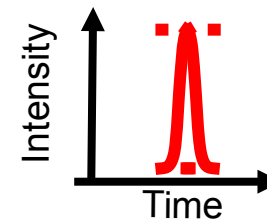
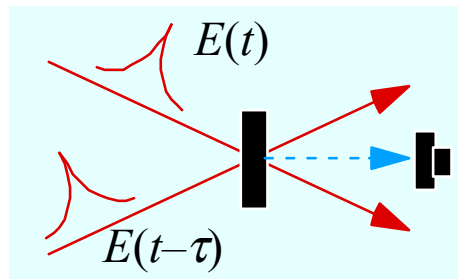
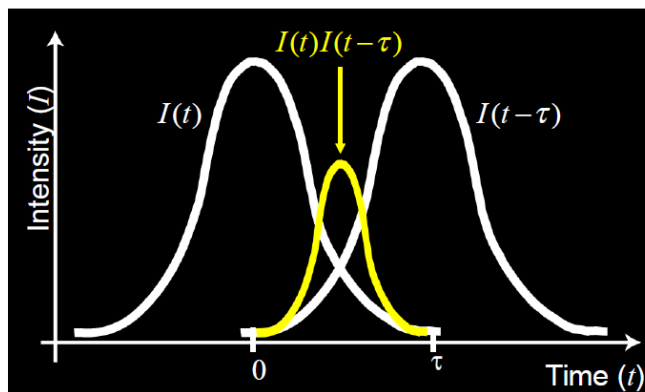
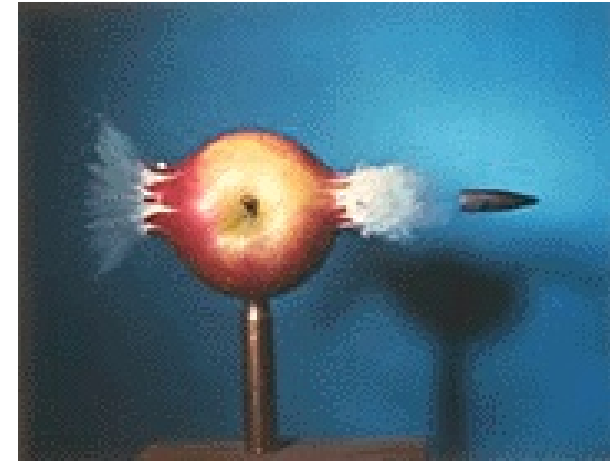
MIT 1964

How can you measure a fs pulse?

Say if you want to take a photograph of this event you need camera having scale of resolution shorter than the event

So If you want to measure fs pulse you need at least fs or shorter probe and so on

The trick is, we use the same pulse to get an estimate



$$A^{(2)}(\tau) \equiv \int_{-\infty}^{\infty} I(t)I(t-\tau) dt$$

**Auto correlation
does not give
phase
information**

To define an ultrashort laser pulse's **intensity** and **phase** vs. time or frequency

Neglecting the spatial dependence for now

$$E(t) = \text{Re} \left\{ \sqrt{I(t)} \exp[i(\omega_0 t - \phi(t))] \right\}$$

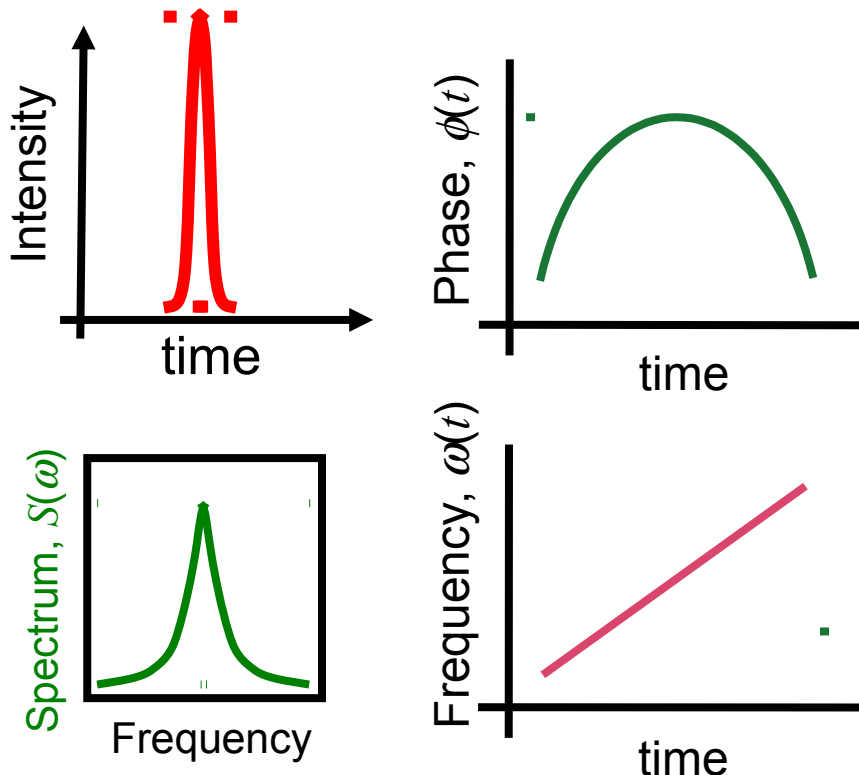
Intensity

Phase

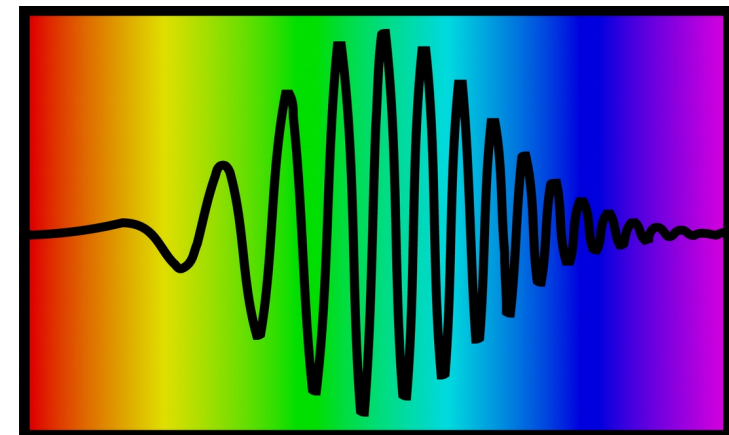
The phase determines the pulse's frequency (i.e., color) vs. time.

$$\omega(t) = \omega_0 - d\phi / dt$$

The instantaneous frequency



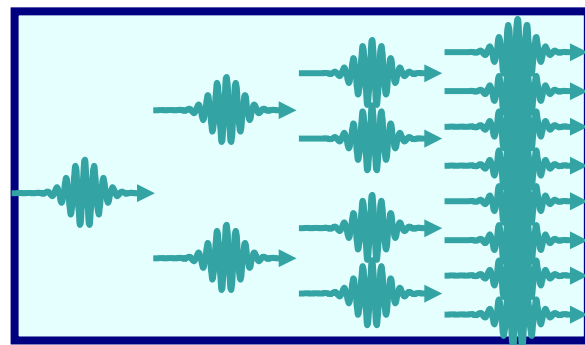
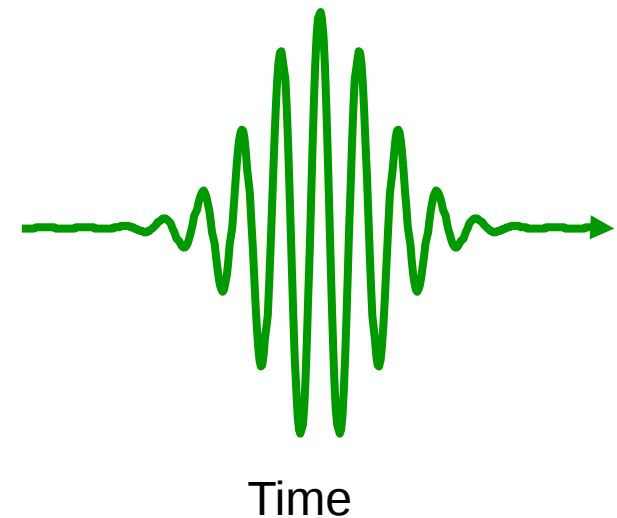
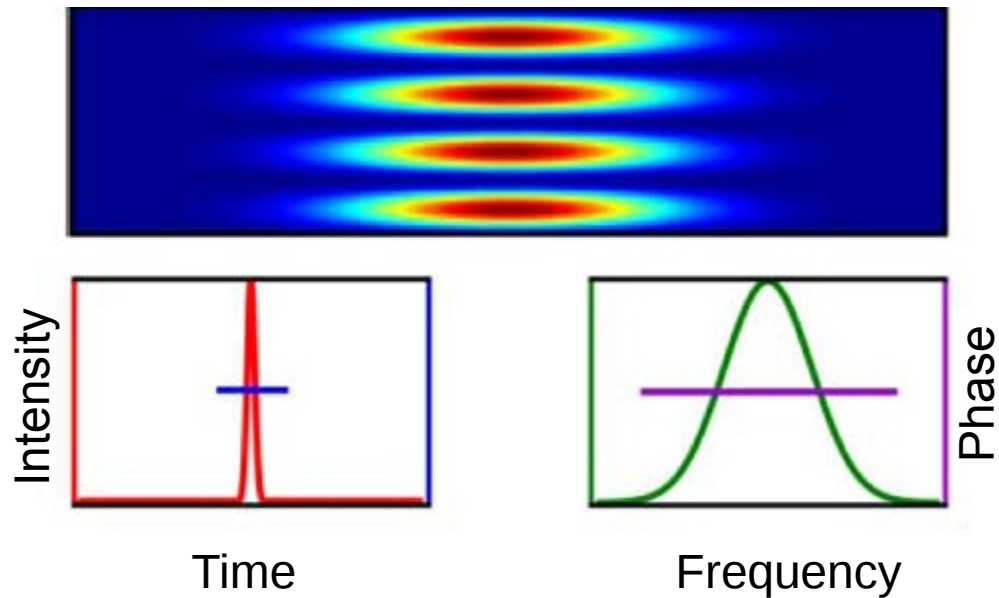
Light electric field



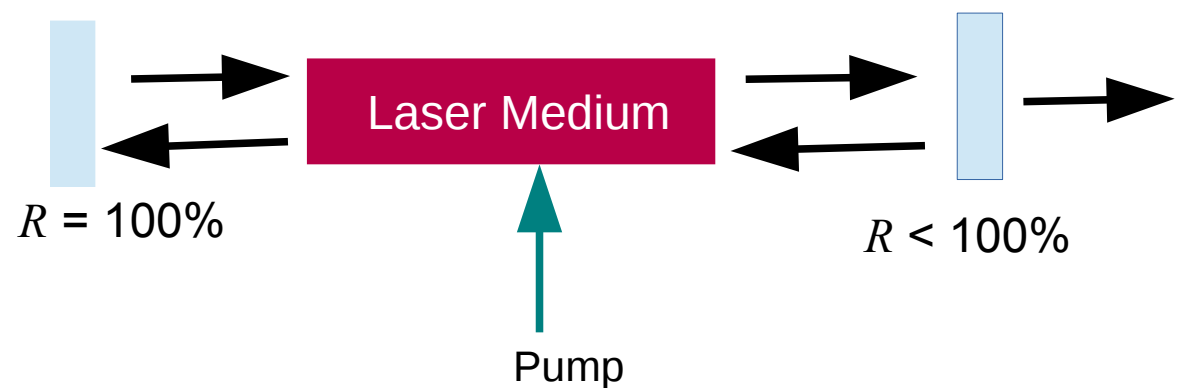
Time

Linearly chirped pulse

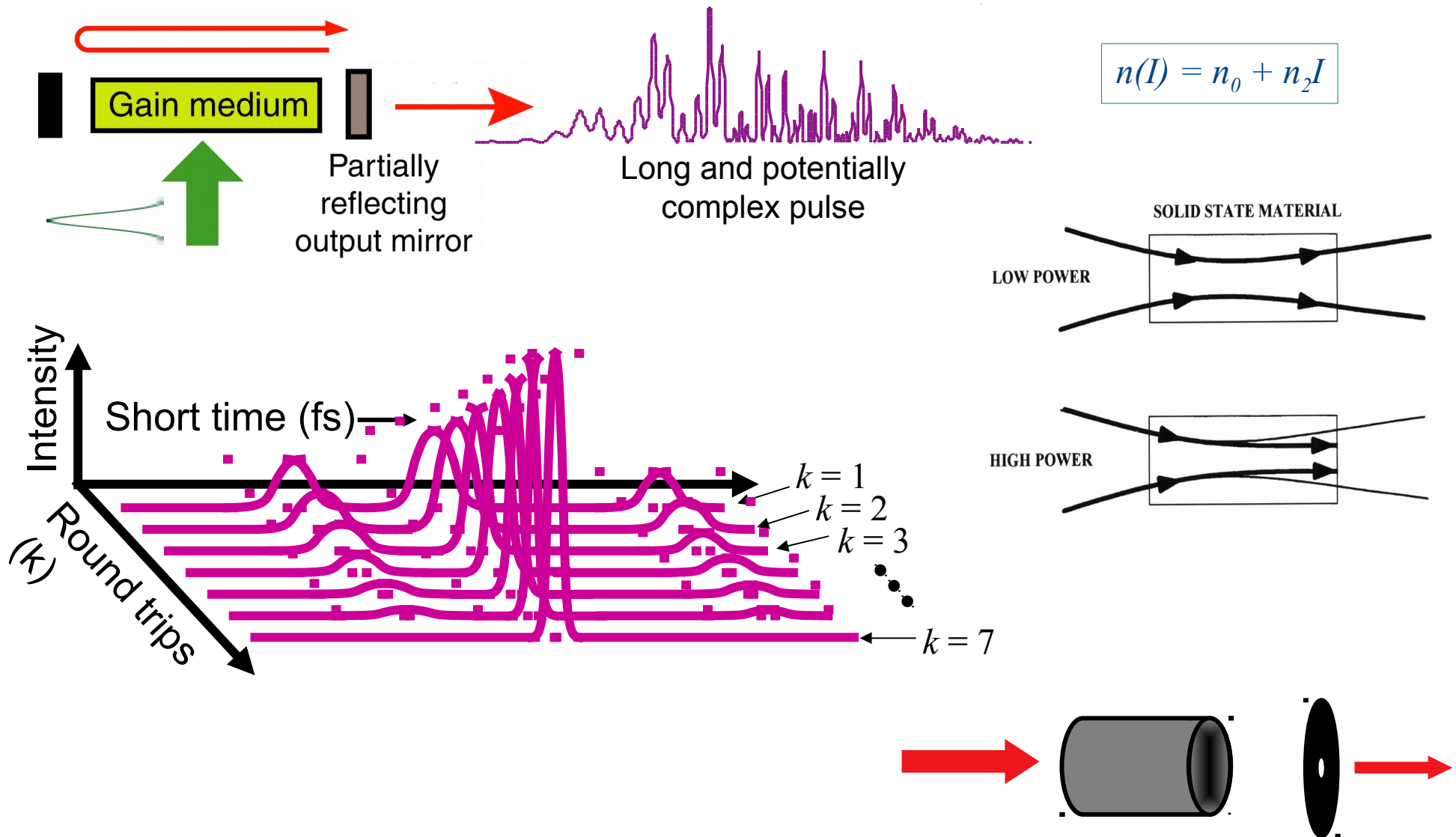
Most of the experiment we need stable Gaussian pulse



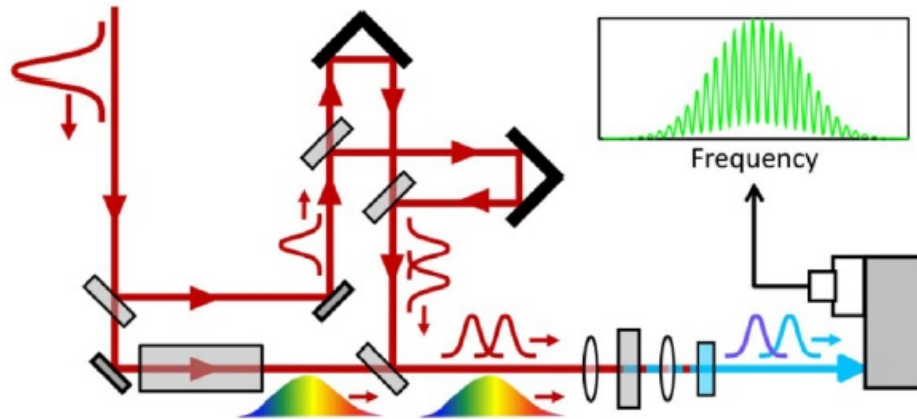
Amplified stimulated emission



fs pulse generation through Kerr-lens mode-locking



Characterize a Pulse before Coherent Spectroscopy



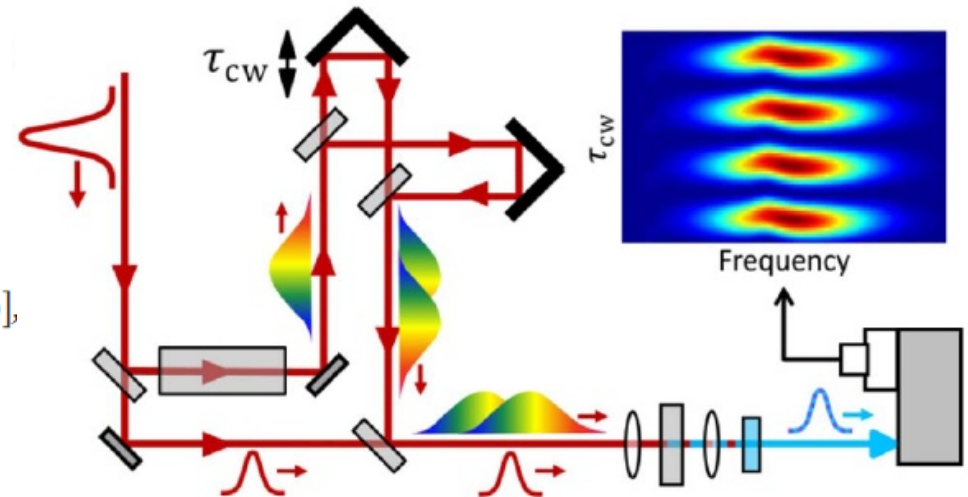
$$S_{\text{SPIDER}}(\omega) = S(\omega) + S(\omega - \Omega) + 2\sqrt{S(\omega)S(\omega - \Omega)} \cos[\omega T + \varphi(\omega) - \varphi(\omega - \Omega)]$$

SPIDER

Spectral Phase Interferometry for Direct Electric-field Reconstruction

$$S_{2\text{DSI}}(\omega, \tau_{\text{cw}}) = S(\omega) + S(\omega - \Omega) + 2\sqrt{S(\omega)S(\omega - \Omega)} \cos[\omega \tau_{\text{cw}} + \varphi(\omega) - \varphi(\omega - \Omega)],$$

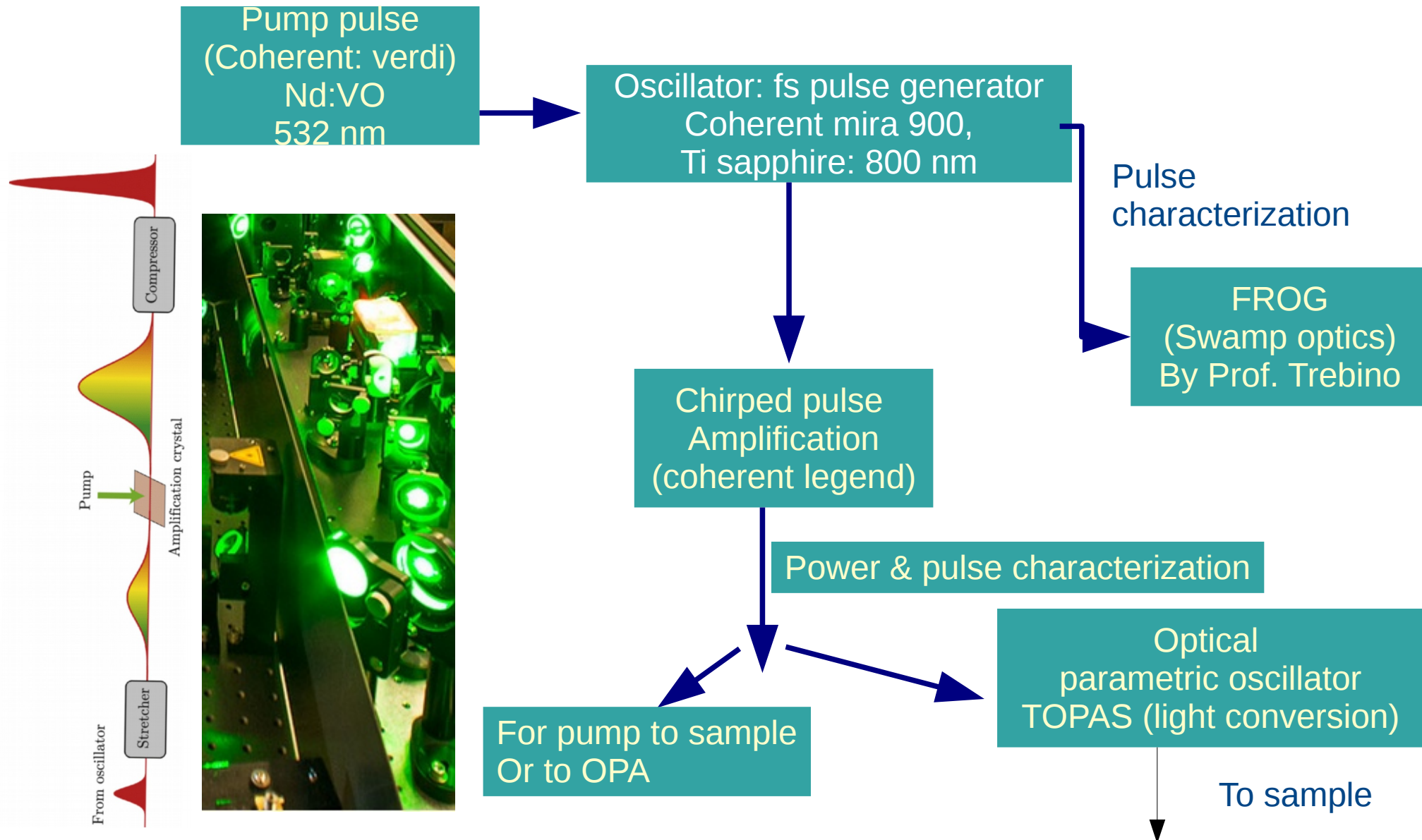
2DSI



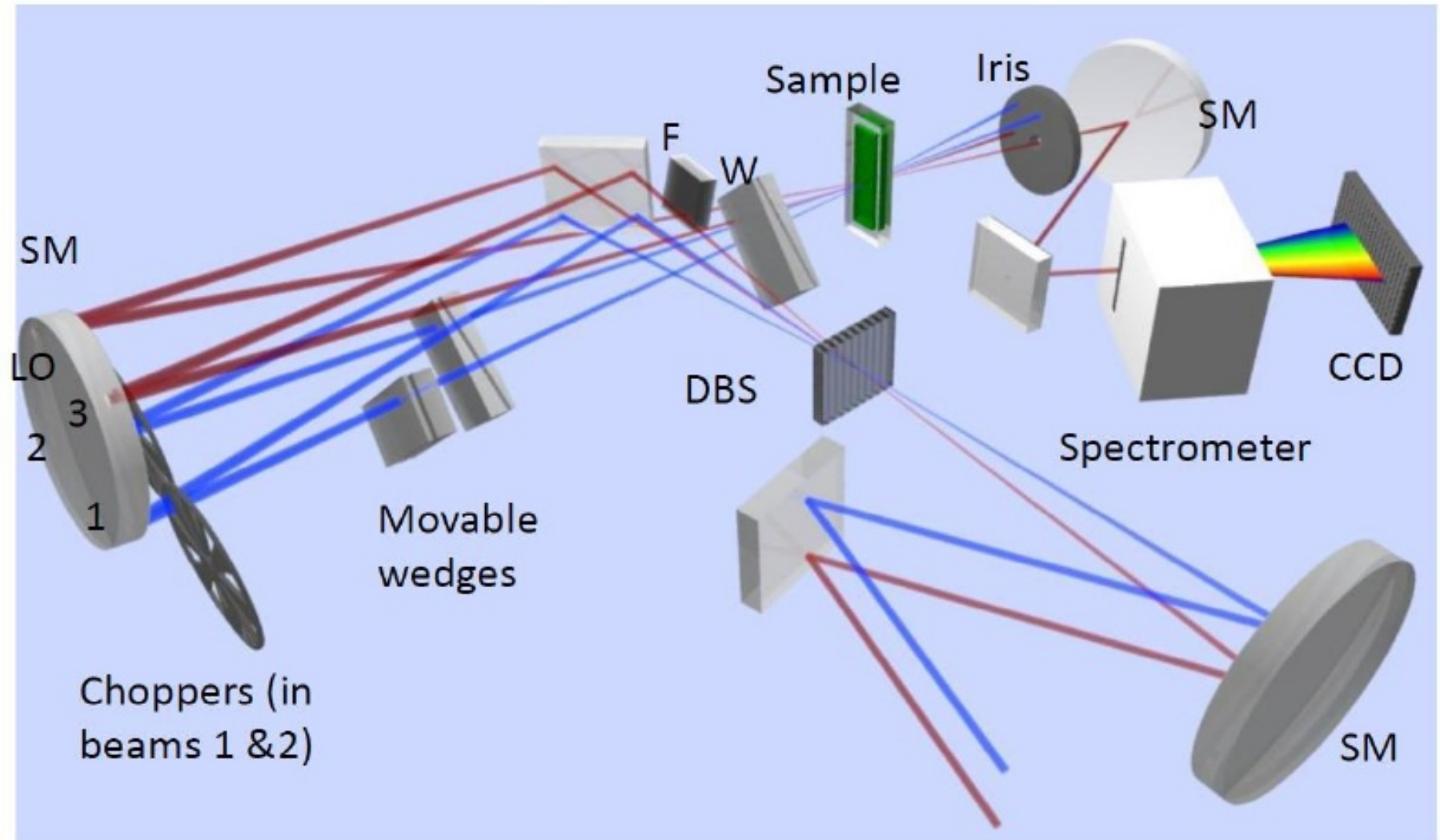
Interferometric methods are inefficient to measure the pulse shape instability

Frequency resolved optical grating (FROG) is more reliable to identify pulse shape instability

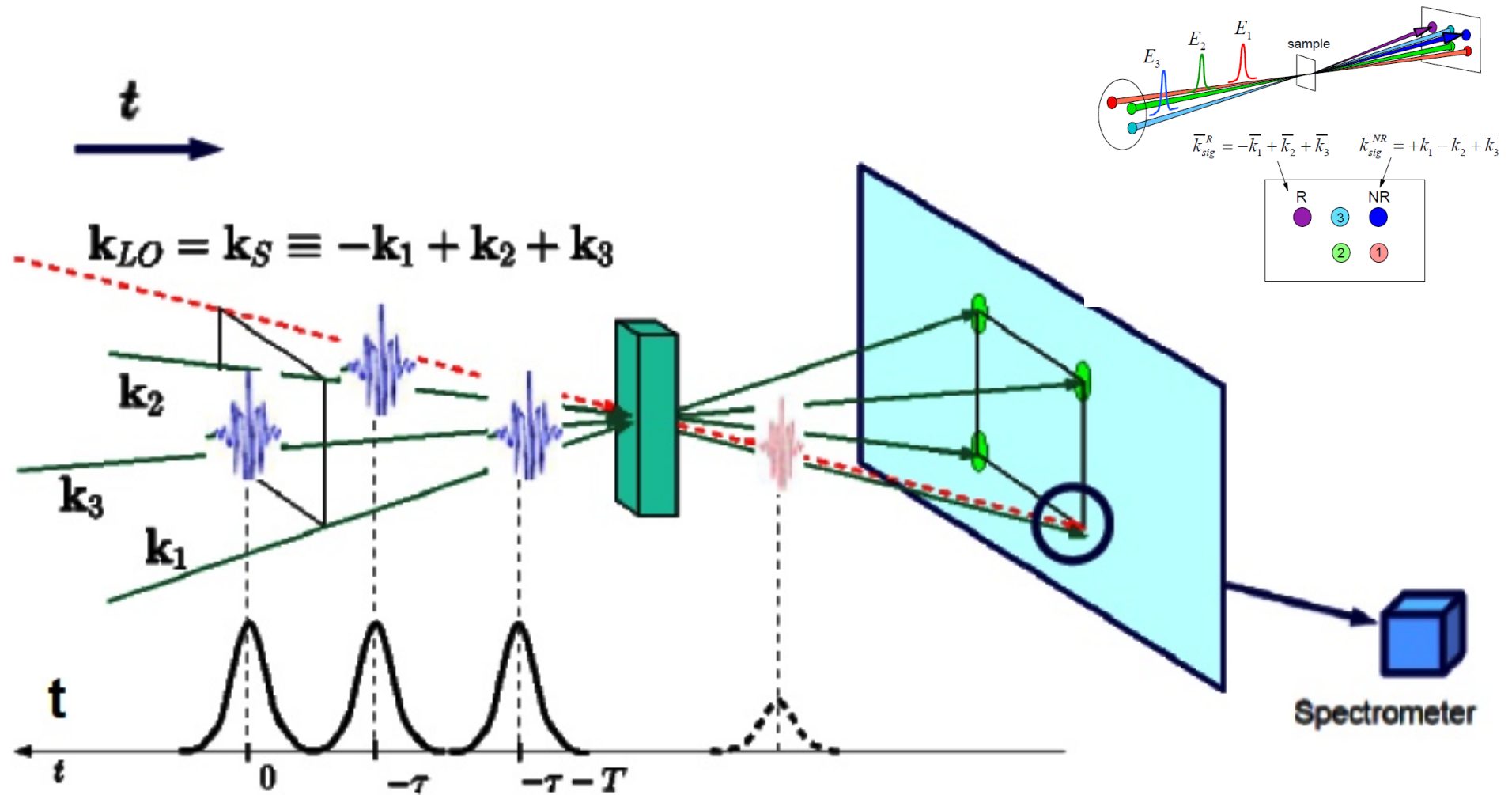
Schematic of the basic set up in a laser lab



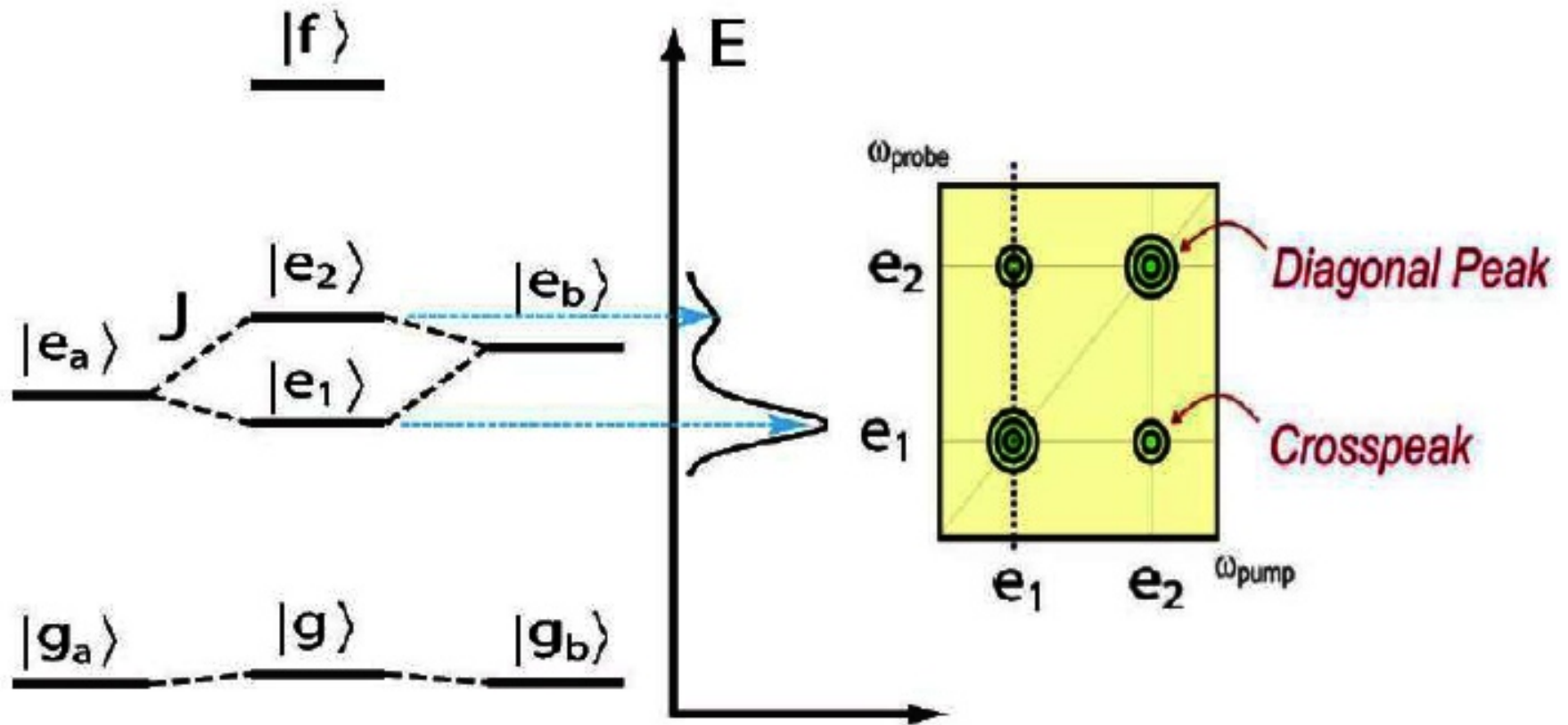
Schematic for coherent spectroscopy



Four wave mixing & Phase matching



Identification of the coupling through coherent spectroscopy



Resolution in the extra frequency dimension allowed detailed studies of excitation dynamics

Underlying Theory

System polarization dynamics

$$\nabla^2 \mathbf{E}(r, t) - \frac{1}{c^2} \frac{\partial^2 \mathbf{E}(r, t)}{\partial t^2} = \mu_0 \frac{\partial^2 \mathbf{P}(r, t)}{\partial t^2} = \mu_0 N q \frac{\partial^2 \mathbf{x}_q(r, t)}{\partial t^2}$$

Electric displacement is $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$

$$\mathbf{P} = \epsilon_0 \left(\chi^{(1)} \cdot \mathbf{E} + \chi^{(2)} \cdot \mathbf{E} \cdot \mathbf{E} + \chi^{(3)} \cdot \mathbf{E} \cdot \mathbf{E} \cdot \mathbf{E} + \dots \right)$$

We measure macroscopic polarization of an ensemble hence need density matrix description

$$\rho(t) = \sum_k p_k |\Psi_k(t)\rangle \langle \Psi_k(t)|$$
$$\langle \mu \rho(t) \rangle = \left\langle \begin{pmatrix} 0 & \mu_1 \\ \mu_2 & 0 \end{pmatrix} \begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix} \right\rangle$$

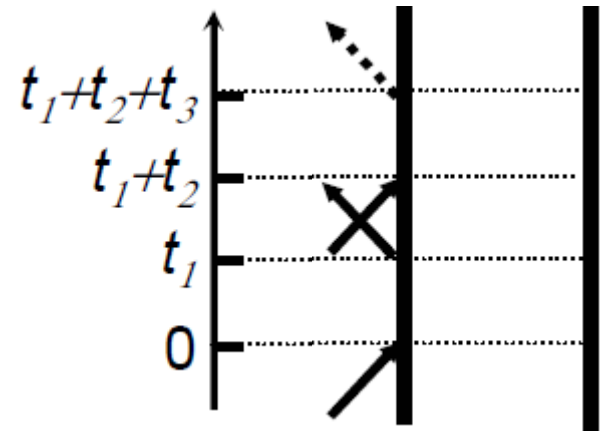
3rd order polarization of the system is given as convolution of 3 electric fields with response function

$$P^{(3)} = \int_0^\infty dt_3 \int_0^\infty dt_2 \int_0^\infty dt_1 E(t - t_3) E(t - t_3 - t_2) E(t - t_3 - t_2 - t_1) S(t_3, t_2, t_1)$$

$$S^{(3)}(t_3, t_2, t_1) = \langle \mu(t_3 + t_2 + t_1) [\mu(t_2 + t_1), [\mu(t_1), [\mu(0), \rho(-\infty)]]] \rangle$$

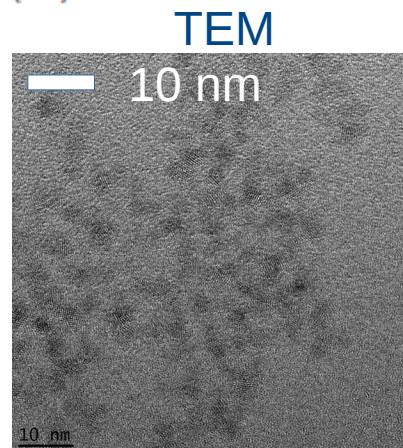
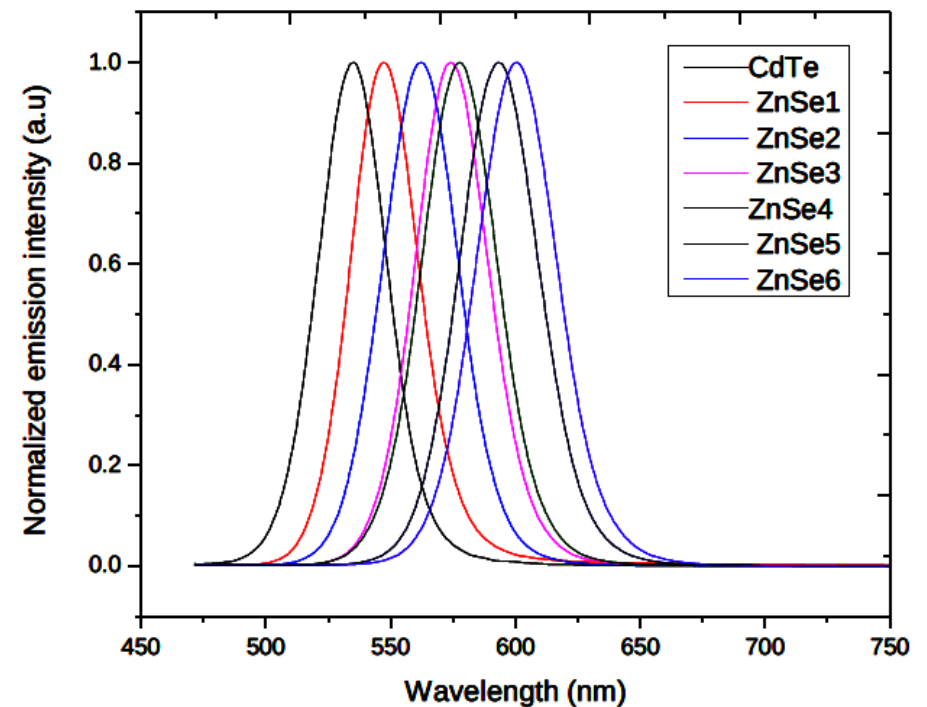
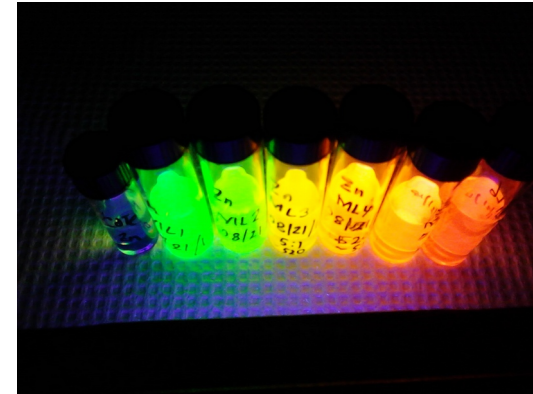
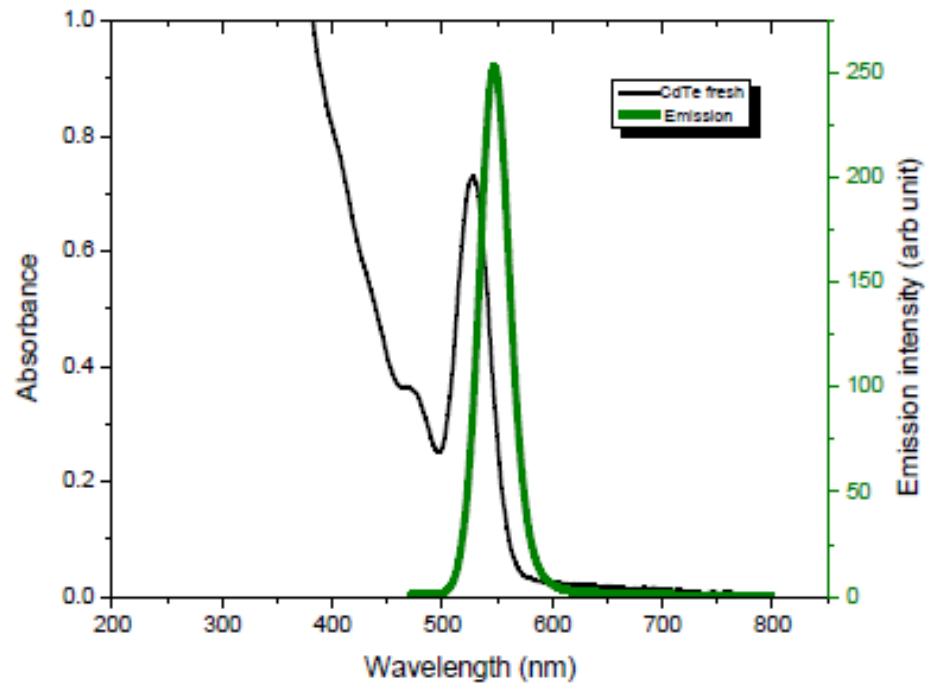
3rd order response function

To get the double Fourier transform spectra

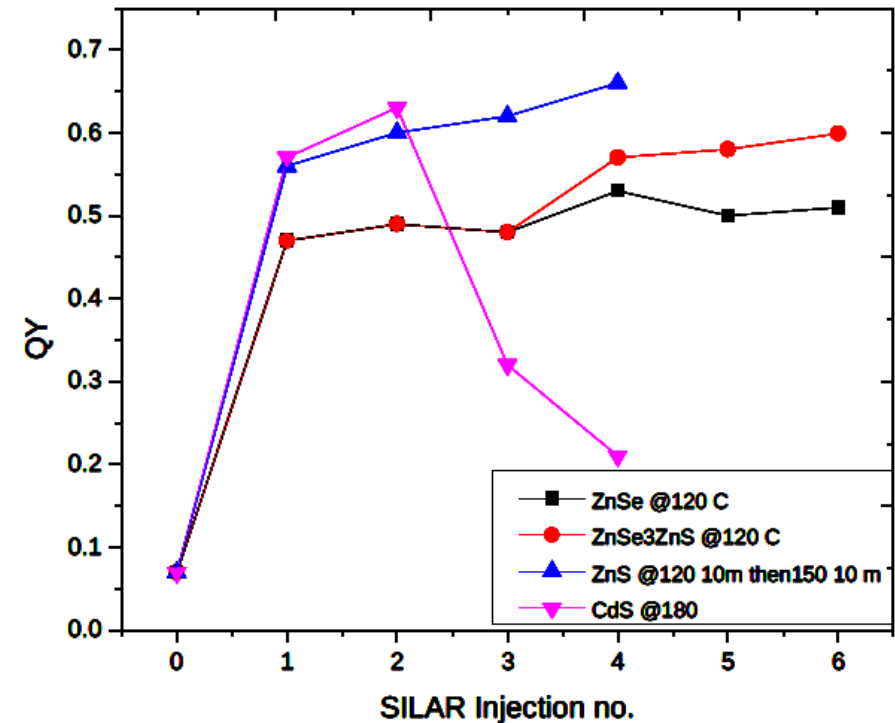
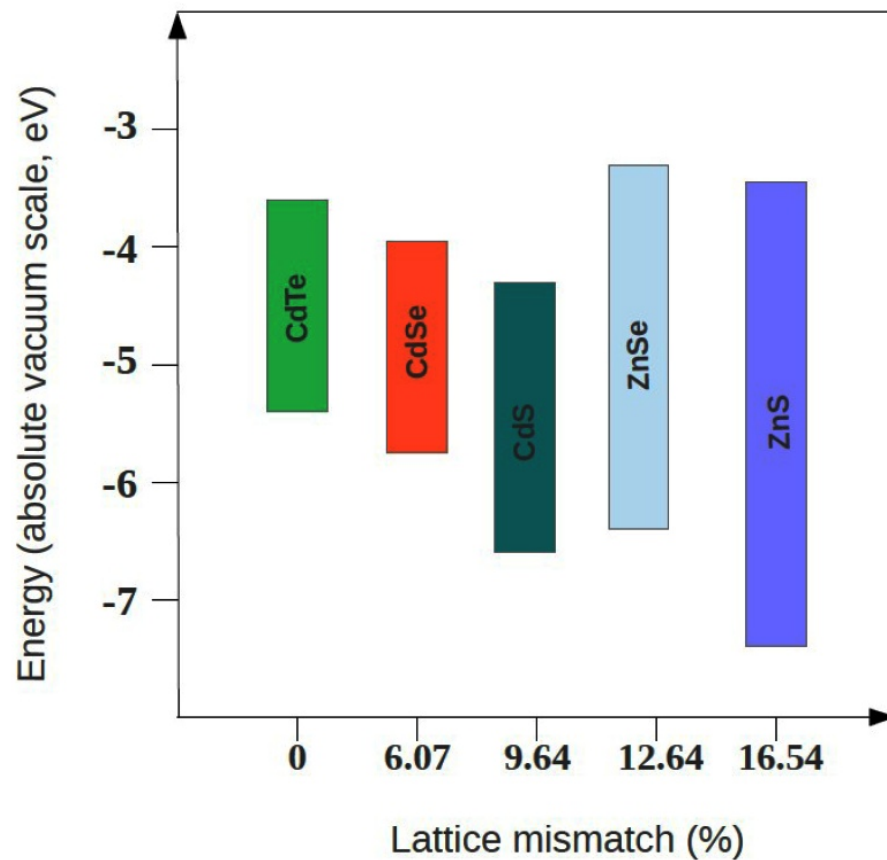


$$S^{(3)}(\omega_3, t_2, \omega_1) = \int_0^\infty \int_0^\infty S^{(3)}(t_3, t_2, t_1) e^{i\omega_3 t_3} e^{\pm i\omega_1 t_1} dt_1 dt_3$$

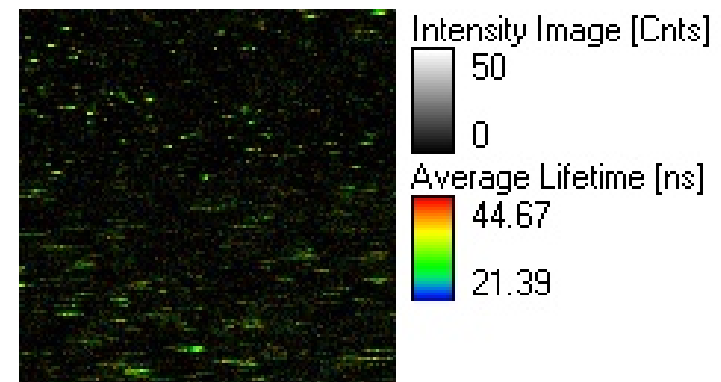
Experimental Studies: Synthesis of QD & Characterizations



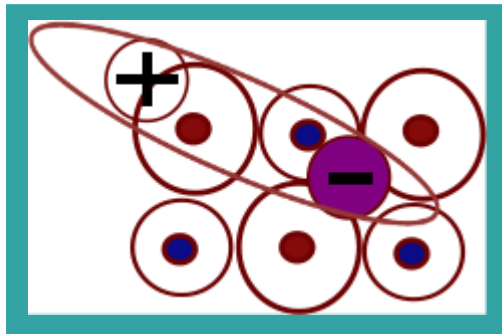
Synthesis of QD & Characterizations



Quantum yield & blinking studies show Type I is brighter than type II QD

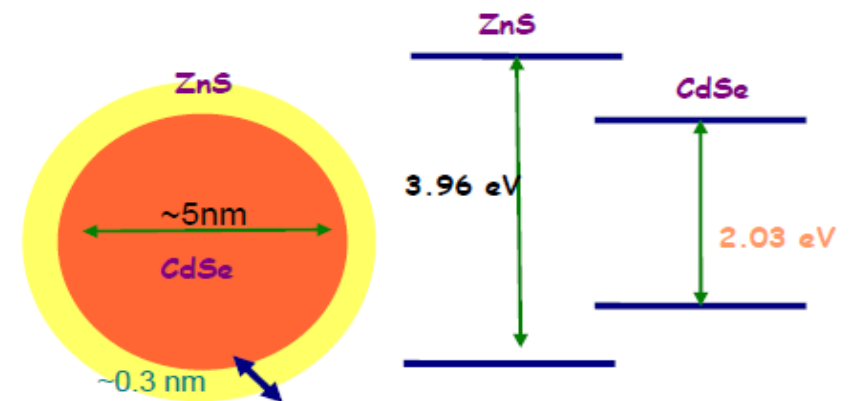
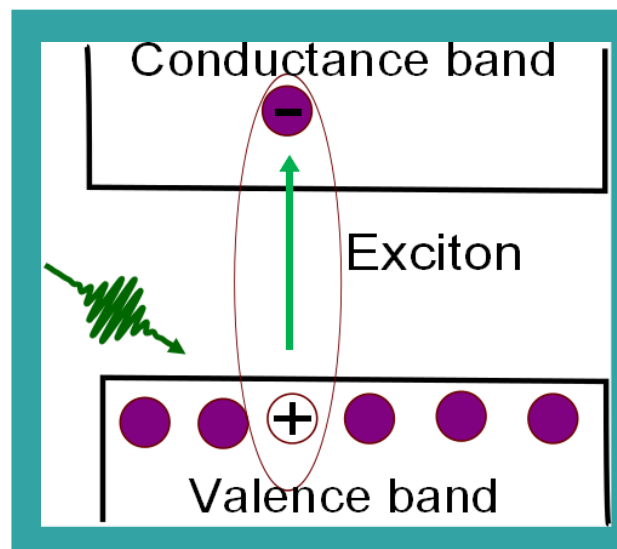


Key observations & system chosen for coherent study



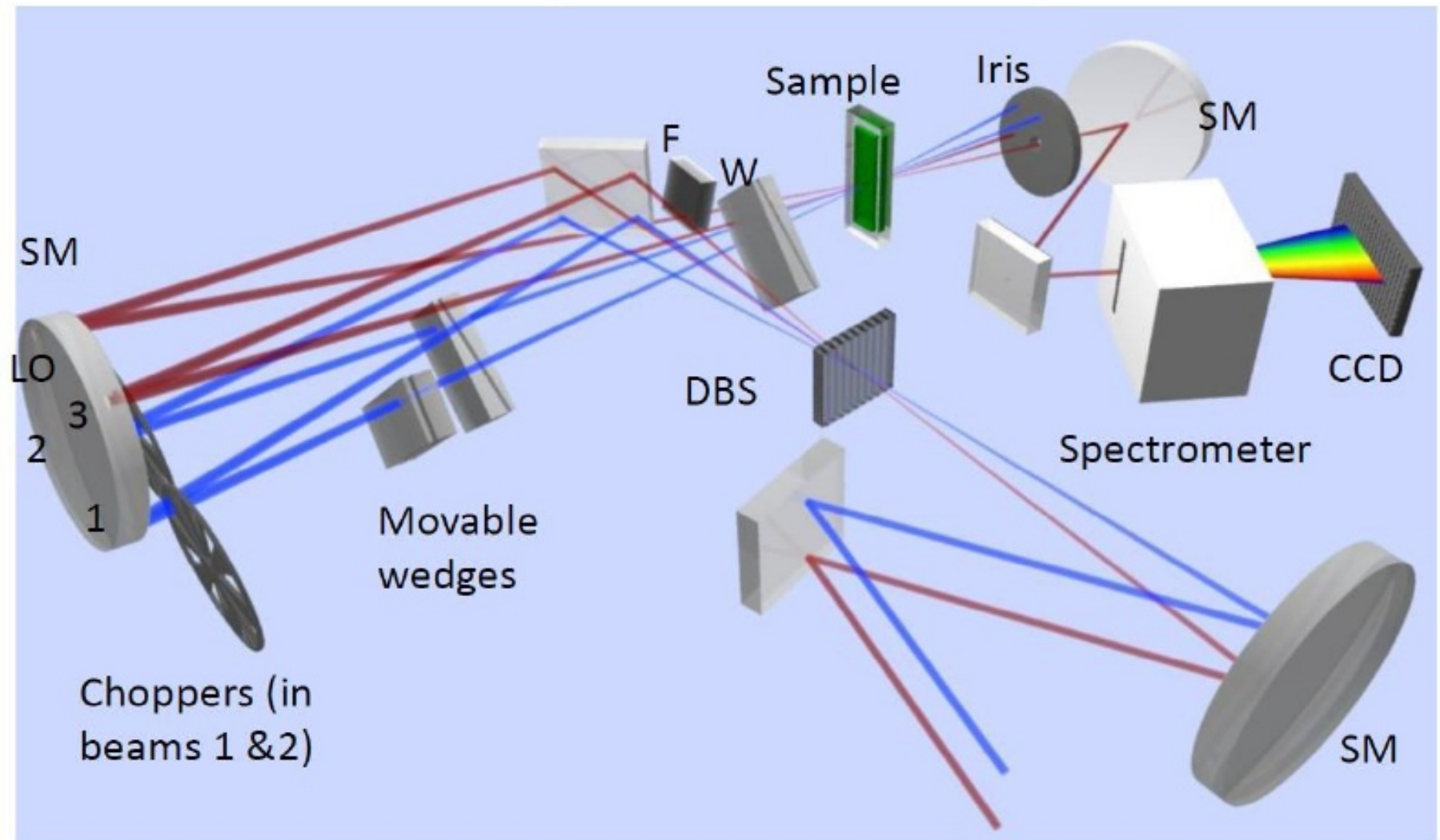
Depending on the exciton energy, lattice matching, stoichiometry, ligand matching to reduce trap states, in general type I core shell quantum dots are more stable and bright, where their radiative recombination rate fast to make them good candidate for photonic application.

Where as in general type II quantum dots shows large charge separation, with slow recombination, promising for photocell applications

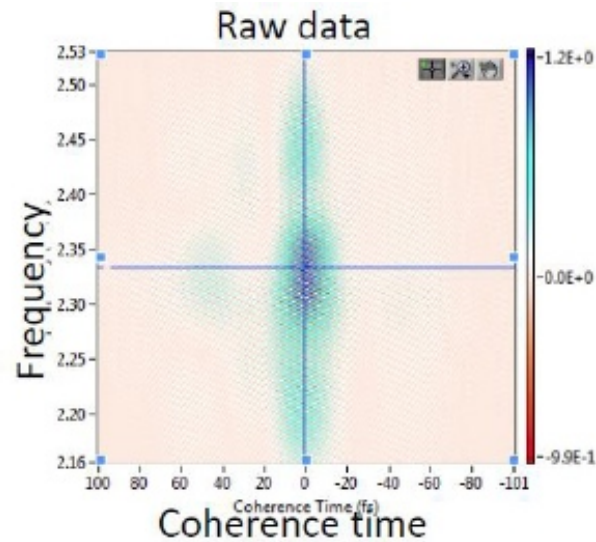


System chosen for coherent study

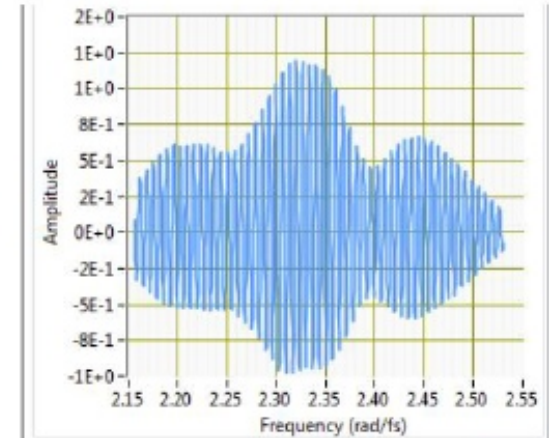
Schematic for coherent spectroscopy



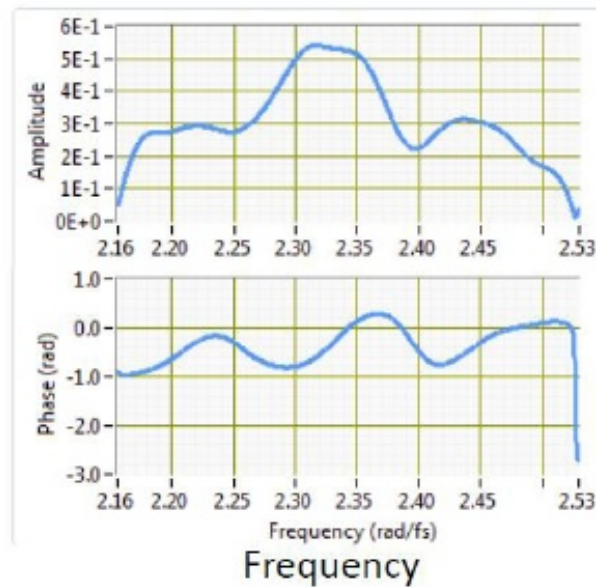
2D data handling



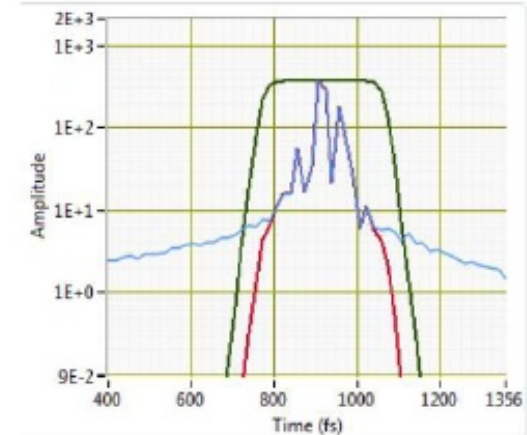
slice spectrum for each τ



FFT (along ω_t)



cut & FFT⁻¹
get $I(\omega)$, $\Phi(\omega)$

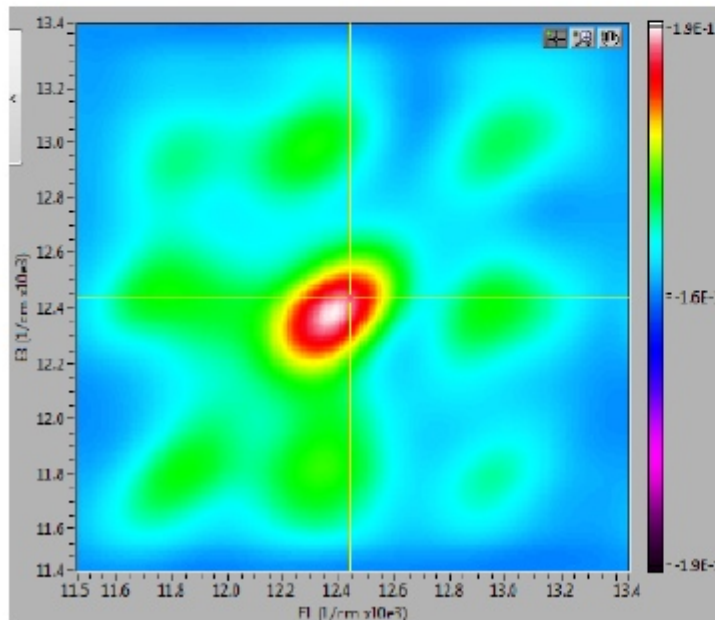


2D Data Analysis

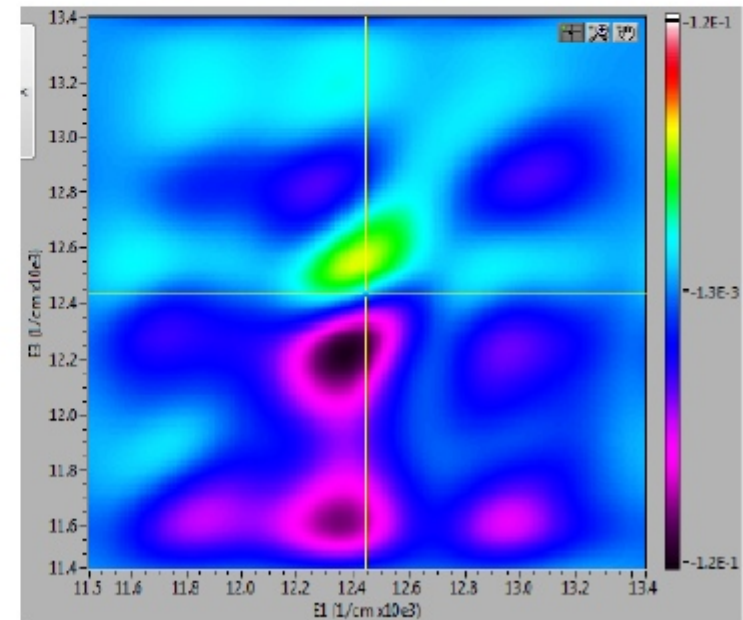
Collect all the τ slices and
 FFT^{-1} along τ

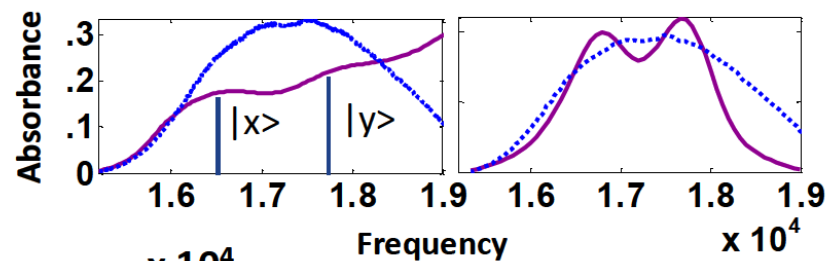


Real 2D spectrum

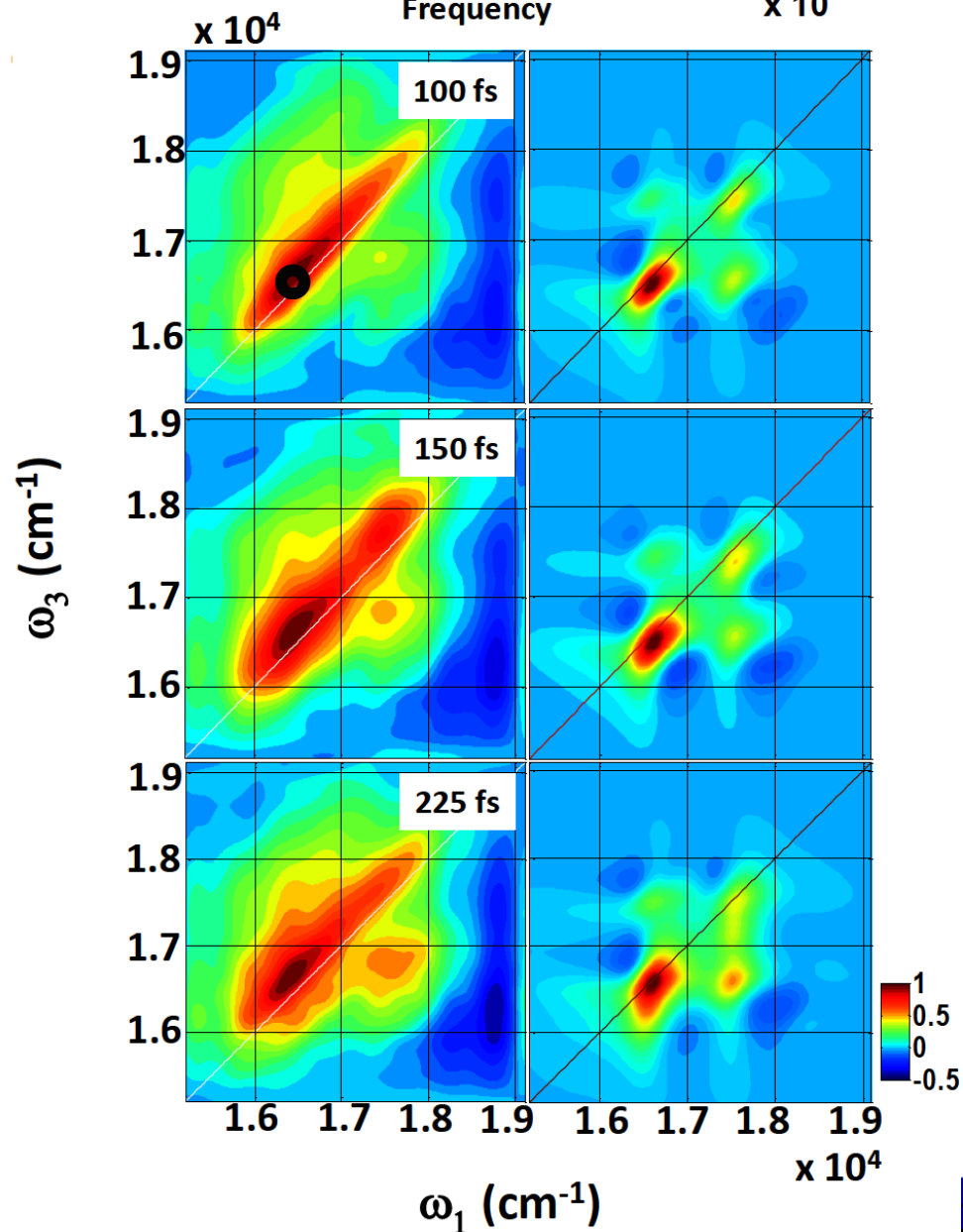


Imaginary 2D spectrum

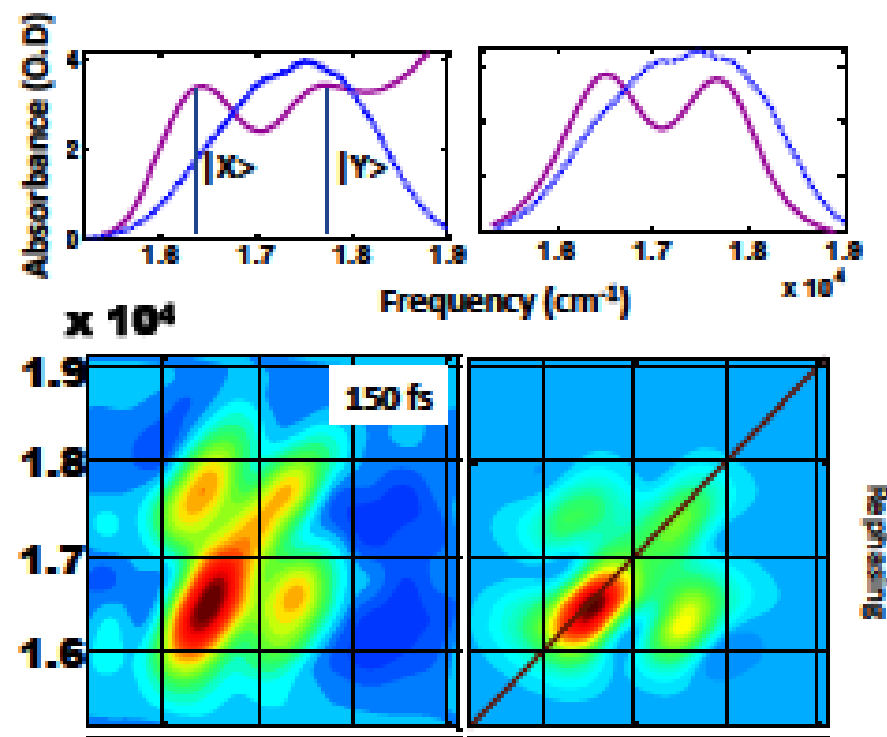




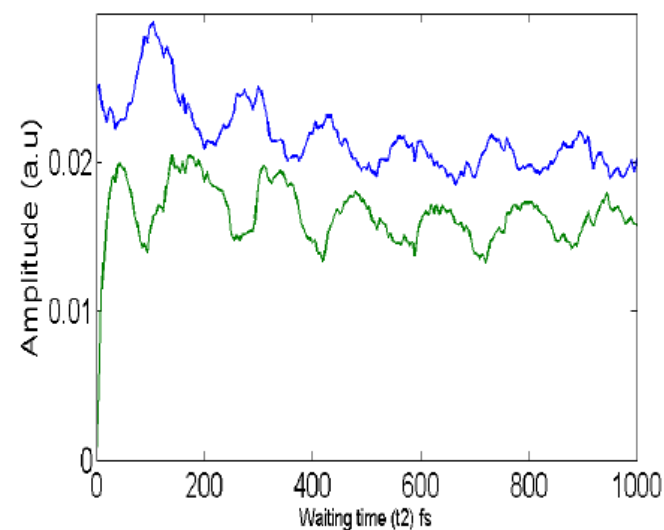
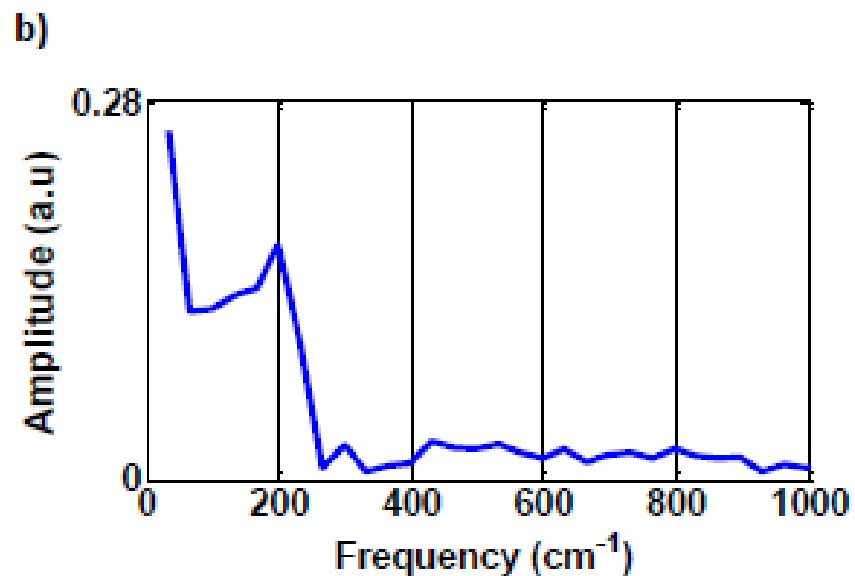
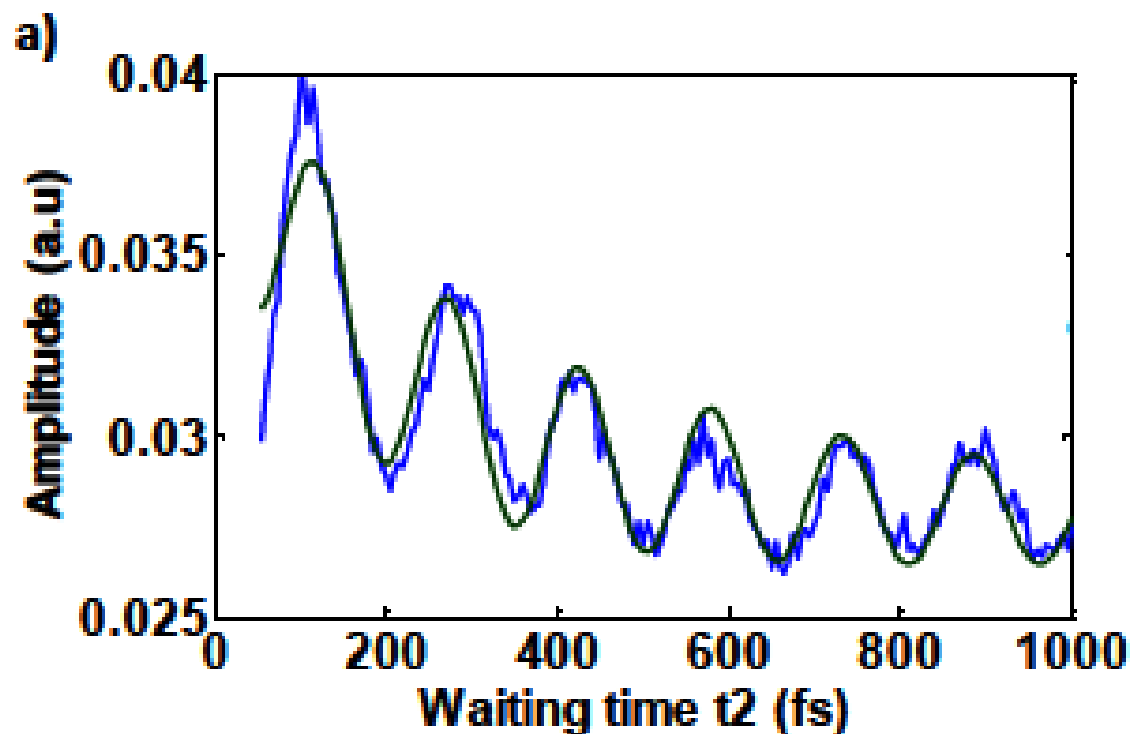
2D electronic spectra of CdSe-ZnS core-shell quantum dot in toluene at different waiting time Experiment and simulation



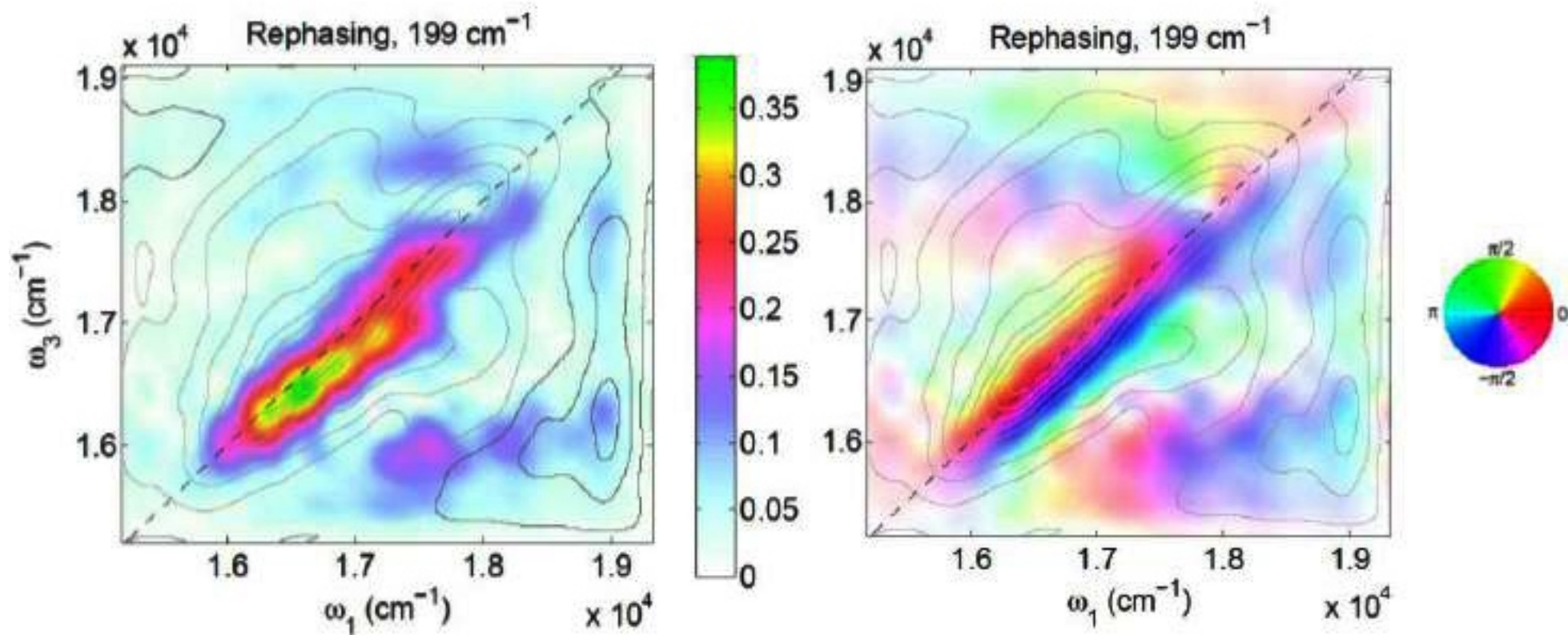
2D electronic spectra CdSe-ZnS core-shell quantum dot in water



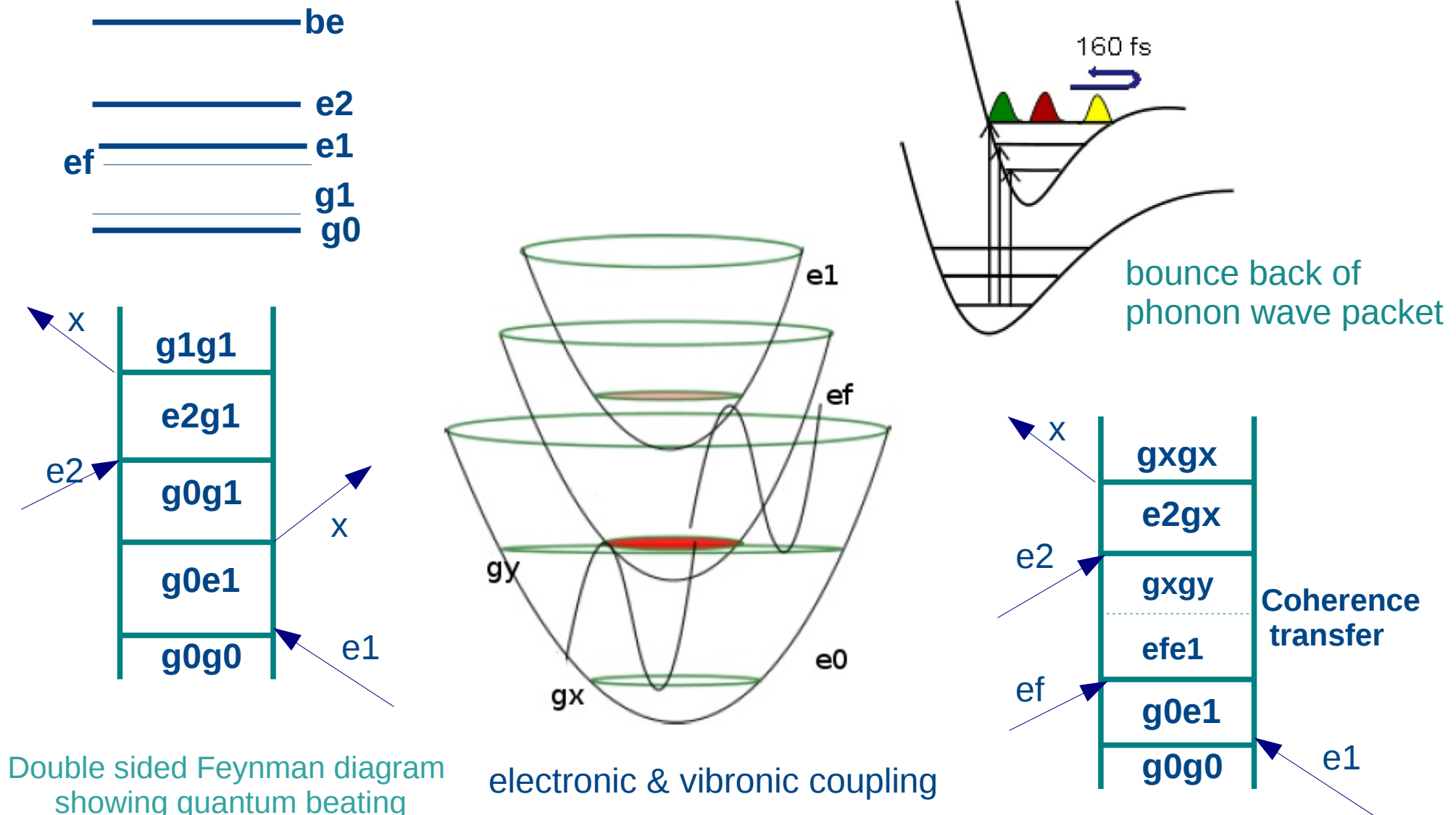
Analysis of Oscillation



Triple Fourier Transform 2DES Showing the Phonon Mode



Phonon wave packet dynamics & possibility of electronic coupling



Summary

We briefly discuss how to create & measure a fs pulse being aware of the possibility of coherent artifact. Where our studies show that even advance 2DSI can measure coherent artifact if proper precautions are not taken.

We see, 2D-optical spectroscopic technique, where the four wave mixing method has a control over the phase information & its basic setup for experiment and the main underlying theory to explain the observed spectra.

How to prepare & characterize Gr IV-VI quantum dots. Where our studies show lattice matching, surface stoichiometry & nature of ligands can play crucial roles. Also depending on the charge separation and nature of charge recombination type I quantum dots in general are brighter and good for photonic applications. Where as type II quantum dots has longer charge separation and promising for photocell applications.

Our coherent studies show:

Specific wave packet dynamics of coherent phonon at room temperature in colloidal quantum dot. Using our new approach of doing third Fourier transform along the waiting time of 2D-OS spectra, to view coherent oscillation in additional dimension, we precisely observe the bounce back of phonon wave packets in 160 fs in the excited electronic potential well. The anti phase phonon oscillation in 2D spectra and the energy matching of the fine electronic state of CdSe with LO phonon also evokes the possibility of electron phonon coupling through coherence transfer.

Theoretical Studies:

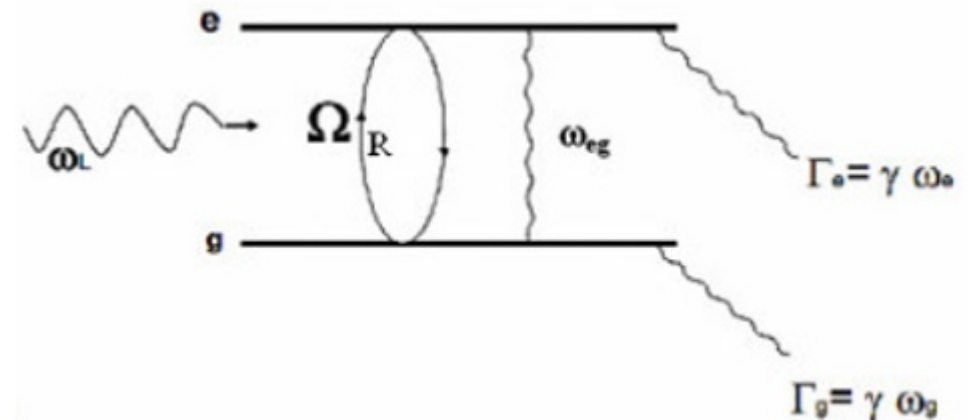
Model study of coherent oscillation in presence of damping in single electron QD

$$m_e \ddot{\vec{r}} + \gamma \dot{\vec{r}} + k \vec{r} = 0$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\vec{r}}} \right) - \frac{\partial L}{\partial \vec{r}} = F_{gen}$$

$$H = \frac{1}{2m_e} \left[(m_e \dot{\vec{r}} + \gamma \vec{r} + \frac{q}{c} \vec{A}) \cdot (m_e \dot{\vec{r}} + \gamma \vec{r} + \frac{q}{c} \vec{A}) \right] + q\phi$$

$$\Gamma_{n,l} = -\gamma(2n + |l| + 1)\Omega$$



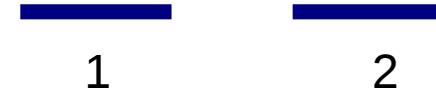
Engineering the coherent superposition through dipolar interactions

$$H = \sum_i n_i \epsilon_i + \sum_{\langle i,j \rangle, \sigma} t_{i,j} (\hat{a}_{i,\sigma}^\dagger \hat{a}_{j,\sigma} + h.c) + \sum_i U_i n_{i\uparrow} n_{i\downarrow} \\ + \sum_{i,j} (V_{i,j}^{ppp} + V_{i,j}^{dipdip}) (n_i - z_i)(n_j - z_j)$$

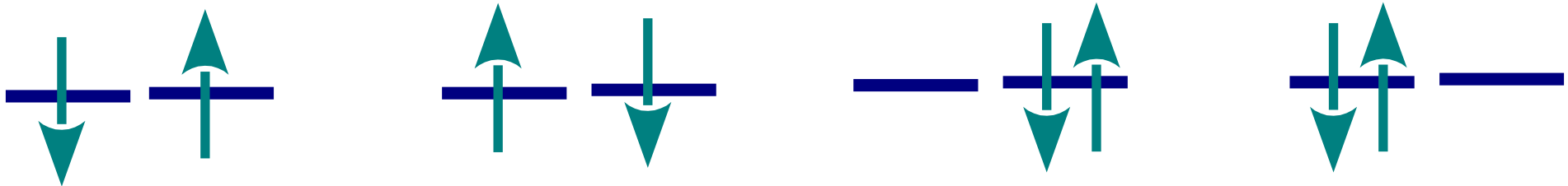
The multi-determinantal CI approach is one of the best methods for obtaining excited-state properties. As the CI is developed in molecular-orbital (MO) space instead of real space, one can have control over the promotions of electrons from the filled MOs to the empty MOs. In principle, a full CI is capable of providing exact solutions for all of the ground and excited states.

Many-body configuration-interaction for excited states

Real space representation of two sites



Possible configuration for singlet state for two sites



$$\sum_i n_{i\uparrow} n_{i\downarrow} = \sum_i \hat{a}_{i,\uparrow}^\dagger \hat{a}_{i,\uparrow} \hat{a}_{i,\downarrow}^\dagger \hat{a}_{i,\downarrow}$$

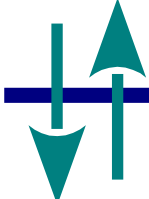
$$\hat{a}_{i,\sigma}^\dagger = \sum_{\mu} a_{\mu,i} \hat{b}_{\mu,\sigma}^\dagger$$

$$H_U = \sum_i \sum_{\mu_1=1}^2 \sum_{\mu_2=1}^2 \sum_{\mu_3=1}^2 \sum_{\mu_4=1}^2 a_{\mu_1,i} a_{\mu_2,i} a_{\mu_3,i} a_{\mu_4,i} (\hat{b}_{i,\uparrow}^\dagger \hat{b}_{i,\uparrow} \hat{b}_{i,\downarrow}^\dagger \hat{b}_{i,\downarrow})$$

We consider the possible configuration as the basis and take their linear combinations
To describe the system in orbital space.

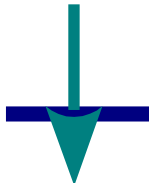
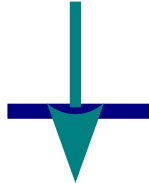
Many-body configuration-interaction for excited states

For coding:

Site	Binary	Decimal
	0	0
	1	1
	10	2
	11	3

For Single site

Degrees of freedom is 4

		01 10	0
		10 10	1
		00 11	2
		11 00	3

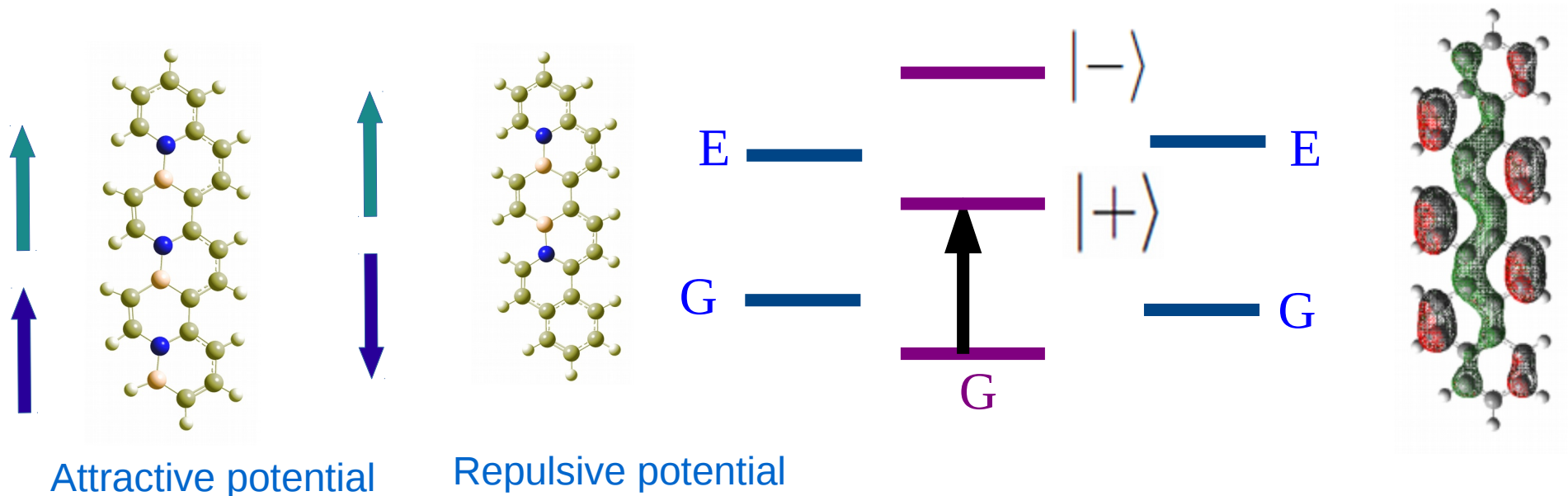
For Two sites degrees of freedom

$2 \times 4 = 16$, Considering singlet spin multiplicity
Possibilities become 4

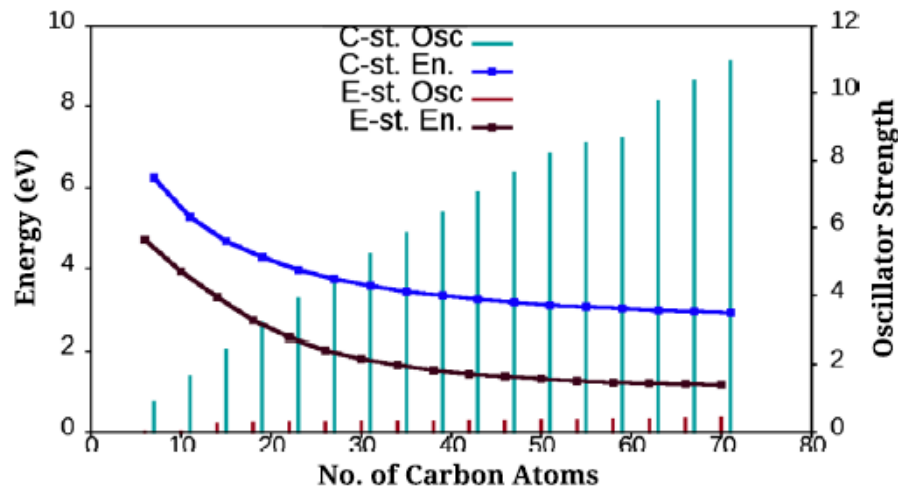
Comparison of CI and DFT

Steps:

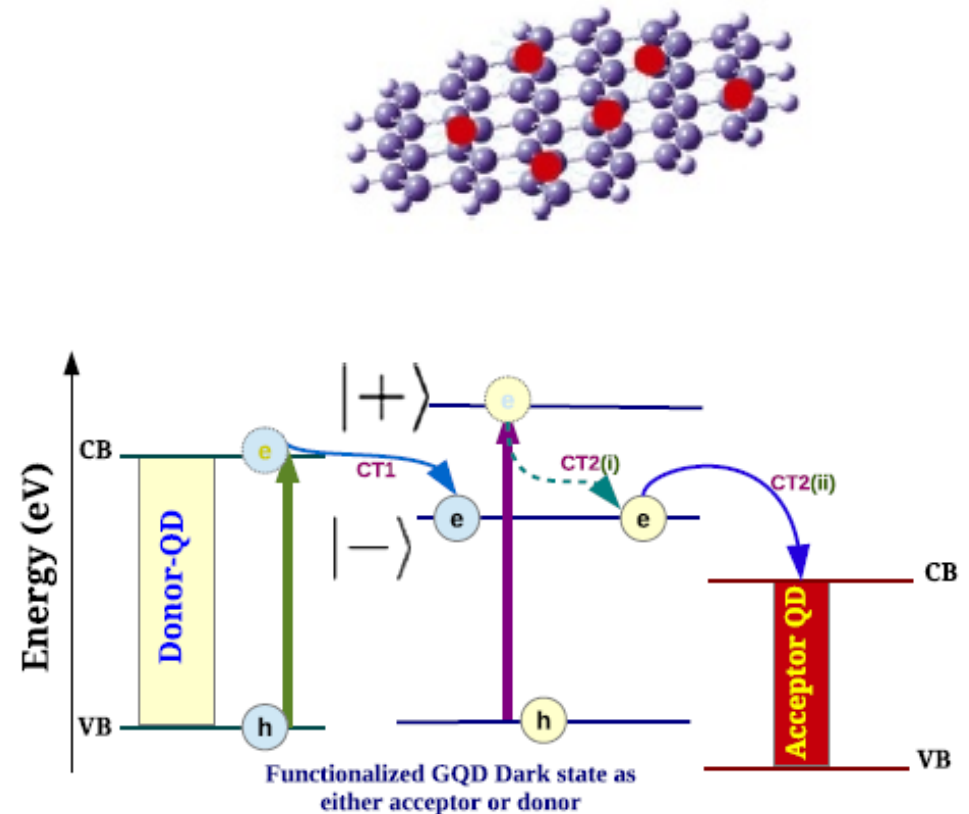
1. Create the tight binding matrix, where correlation is diagonal, hopping is off diagonal
2. diagonalize the tight binding matrix and get the eigen values
3. create the real space basis using binary descriptions
4. transform the real space basis in to orbital representation considering linear coefficient expansion where correlation terms are off diagonal, and tight binding energies are diagonal
5. diagonalize the Hamiltonian matrix and get all the energy values and the eigenvectors.



Many-body configuration-interaction & first-principles studies on graphene quantum dots

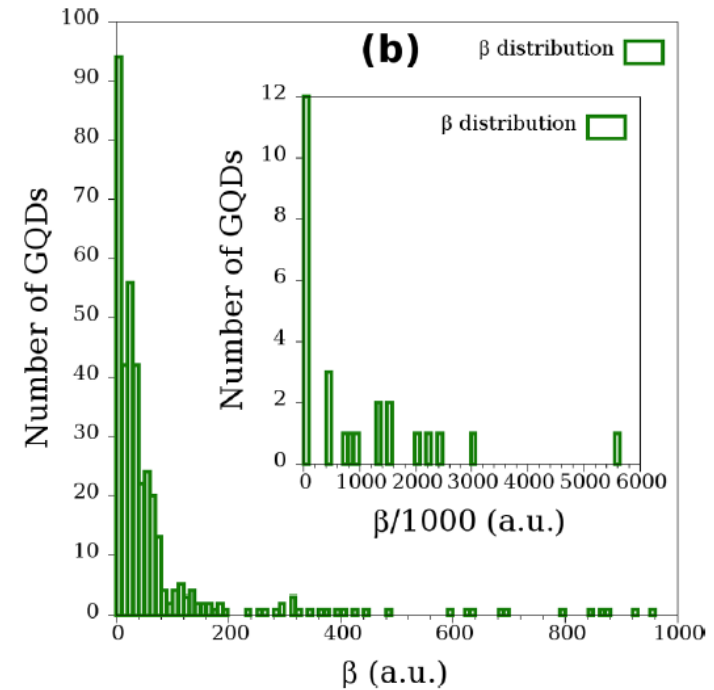
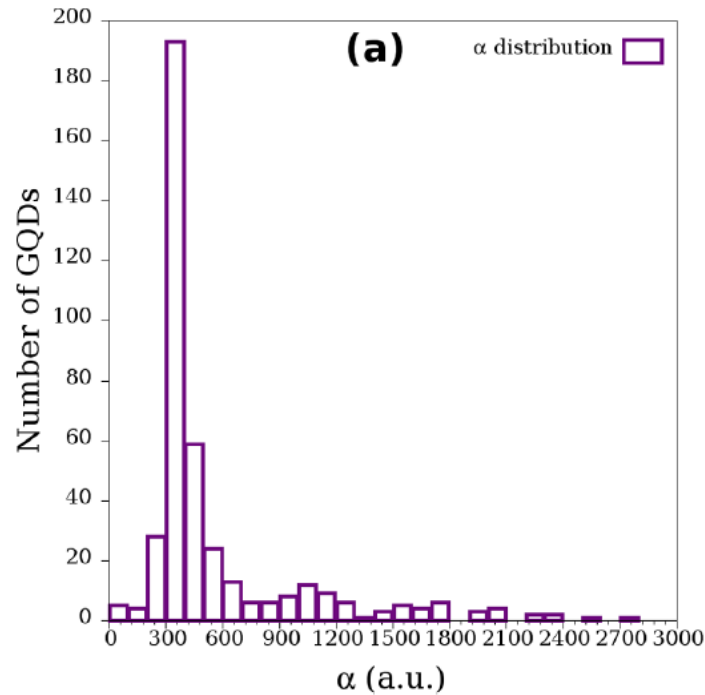


For edge states, the oscillator strengths remain more or less the same but, for the coherent states, the strengths increase steadily



Attractive potential results red end bright states where as repulsive potential gives red end dark states

The spectral shift and trends in the optical nonlinearity of GQDs depend on their structure, size, and shape

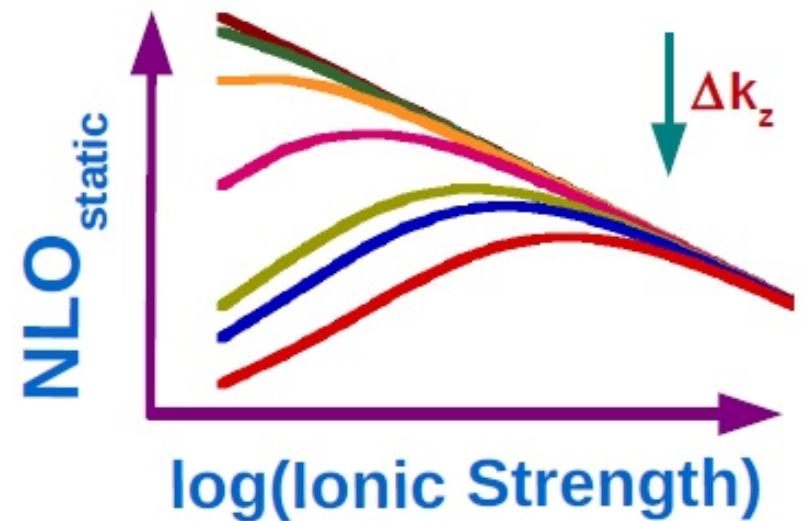
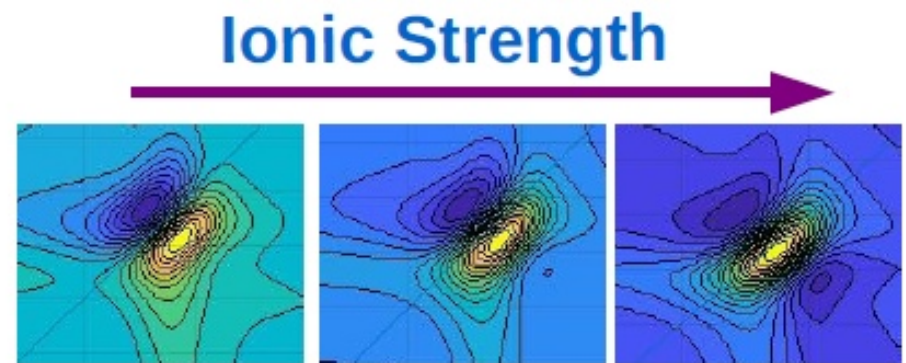
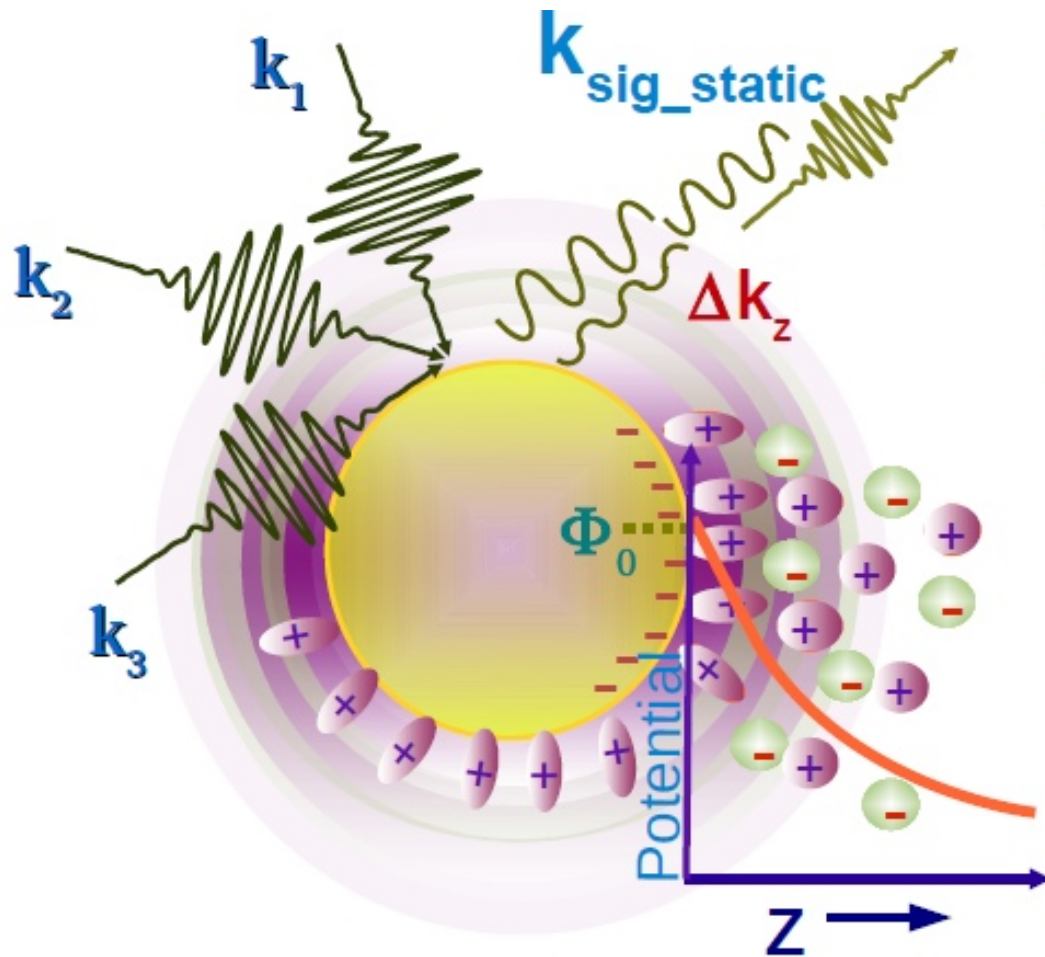


DFT and semi-empirical studies on large varieties of GQD shows that ,
GQDS with less edge state transition energy shows more tendency towards higher nonlinear activity

$$\mu_i = \mu_{0i} + \alpha_{ij}F_j + \frac{1}{2}\beta_{ijk}F_jF_k + \frac{1}{6}\gamma_{ijkl}F_jF_kF_l + \dots \quad (1)$$

$$E(F) = E(0) - \mu_i F_i - \frac{1}{2!}\alpha_{ij}F_iF_j - \frac{1}{3!}\beta_{ijk}F_iF_jF_k - \frac{1}{4!}\gamma_{ijkl}F_iF_jF_kF_l - \dots \quad (2)$$

Effects of Interface on FWM



Conclusion

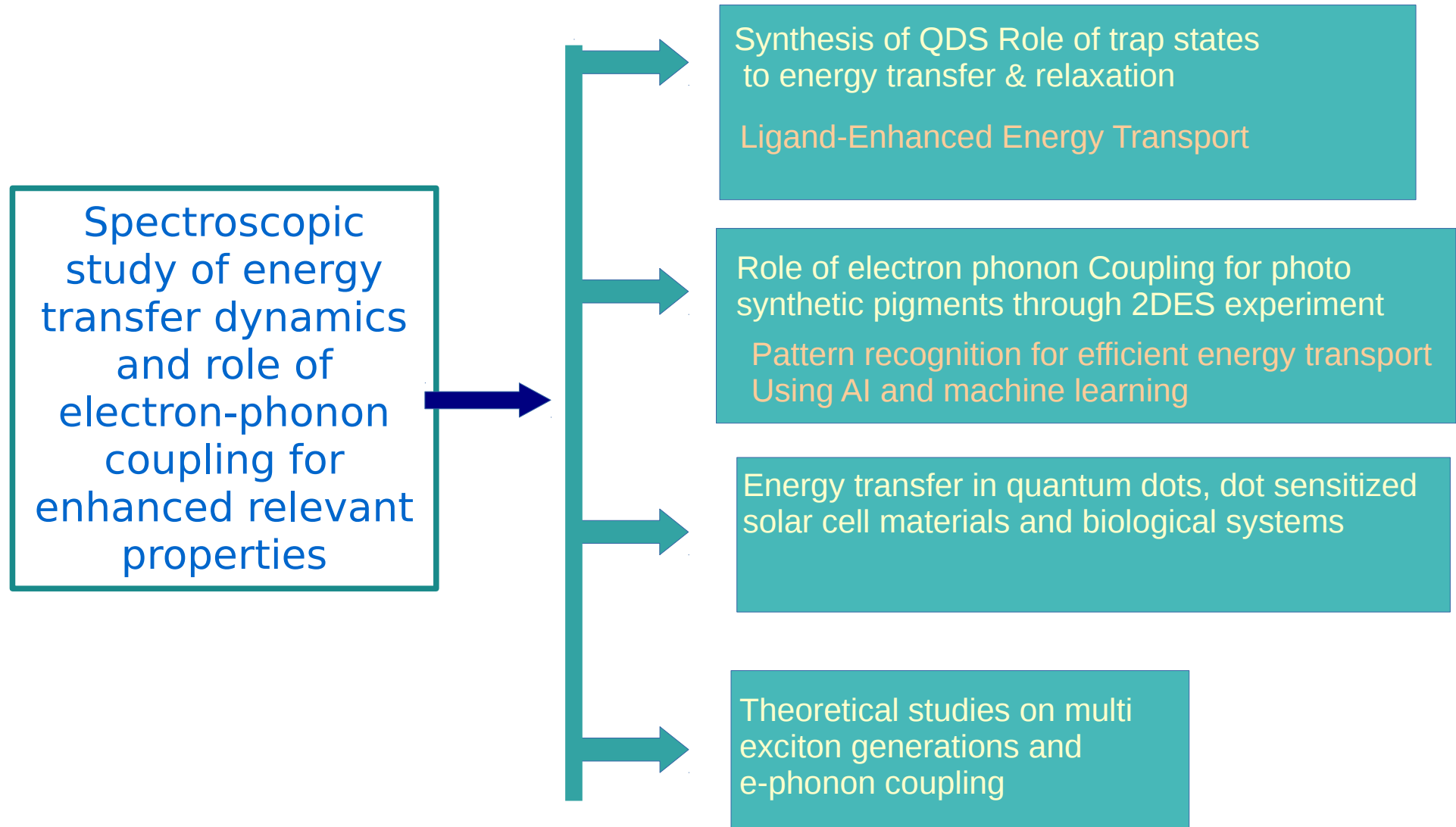
Our coherence study and the nature of electron phonon coupling is one of the fundamental topic of interest to reveal the nuclear motions that occur in electronic excited states. Also rise a question of the contribution of the nested potentials with resonant vibrational level.

Our combined study of experiment & theory provides overall in depth ideas on the nature of the quantum dots, how to control & engineer the optical properties of the quantum dots to make them efficient electron transfer which could be attractive for photonic applications or for photo cell.

References:

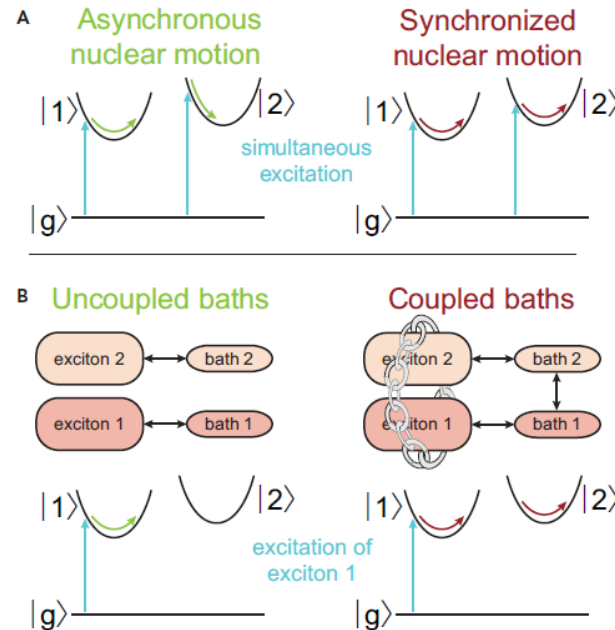
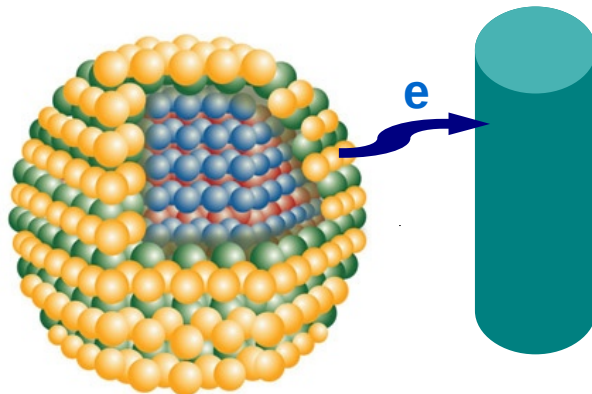
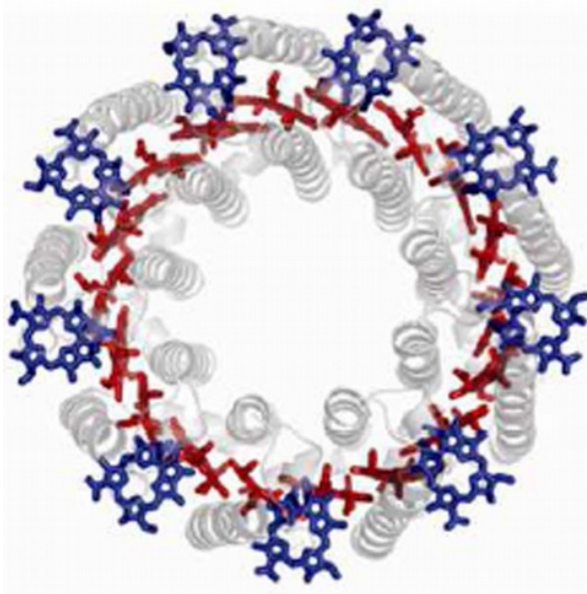
1. **Mukhopadhyay, M.**; Pandey, B.; Pati, S. K. Phys. Rev. App. 2016, 6, 044014
2. **Mukhopadhyay, M.**; Augulis, R.; Lenngren, N.; K.; Zidek, Zigmantas, D.; Pullerits, T. Coherent Multidimensional Spectroscopy, 2012 http://www.mbi-berlin.de/cmds2012/images/program_cmds2012.pdf
3. Yamijala S. SRKC; **Mukhopadhyay, M.**; Pati, S. K. J. Phys. Chem. C, 2015, 119, 12079–12087
4. Rhodes, **M. Mukhopadhyay**, Jonathan Birge, Günter Steinmeyer, Saidur Rahaman, Rick Trebino, Lasers and Electro-Optics (CLEO), 2014, by IEEE
5. **Mukhopadhyay, M.**; Hazra, R.K.; Ghosh, M.; Mukherjee, S.; Bhattacharyya, S.P, Cent. Eur. J. Phys, 2012, 10(4) , 983988.
6. Rhodes, M.; **Mukhopadhyay, M.**; Birge, J.; and Rick Trebino, JOSA B, 2015, 32 (9), 18811888

Future Outlook



Energy transfer dynamics

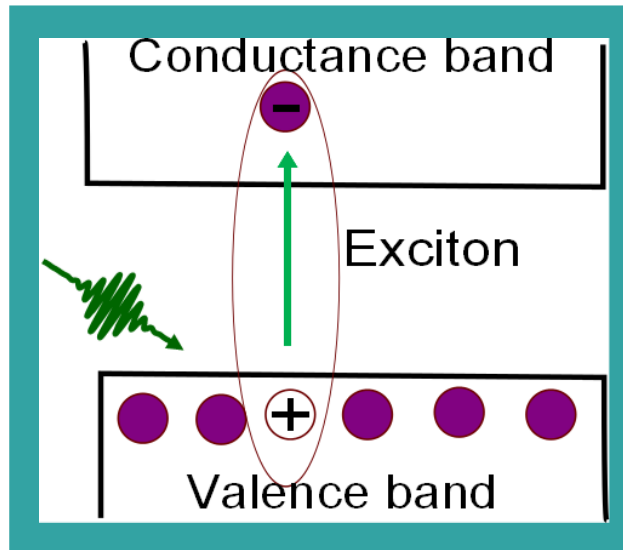
LH2 System



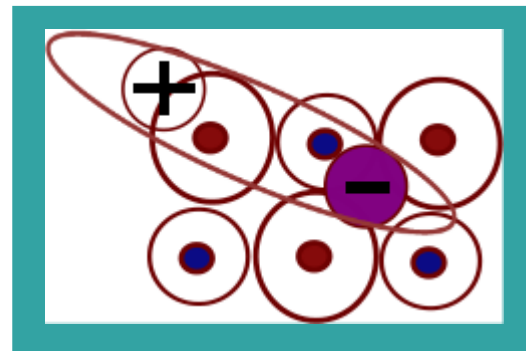
coherent oscillations can originate from (1) pure vibrational states, (2) vibronic (a combination of coupled electronic and vibrational) states, or (3) pure electronic states experiencing correlated nuclear motions

Ultrafast studies on electron transfer dynamics for quantum dot sensitized photocell materials

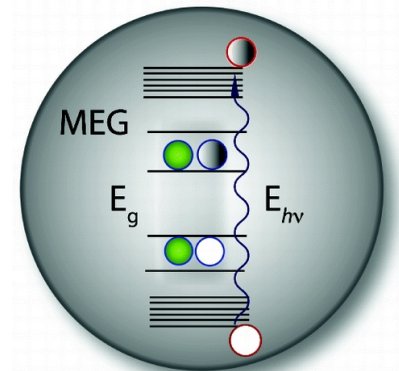
Multi exciton generation : creation of Coherent superposition of several electron-hole pairs by a single photon



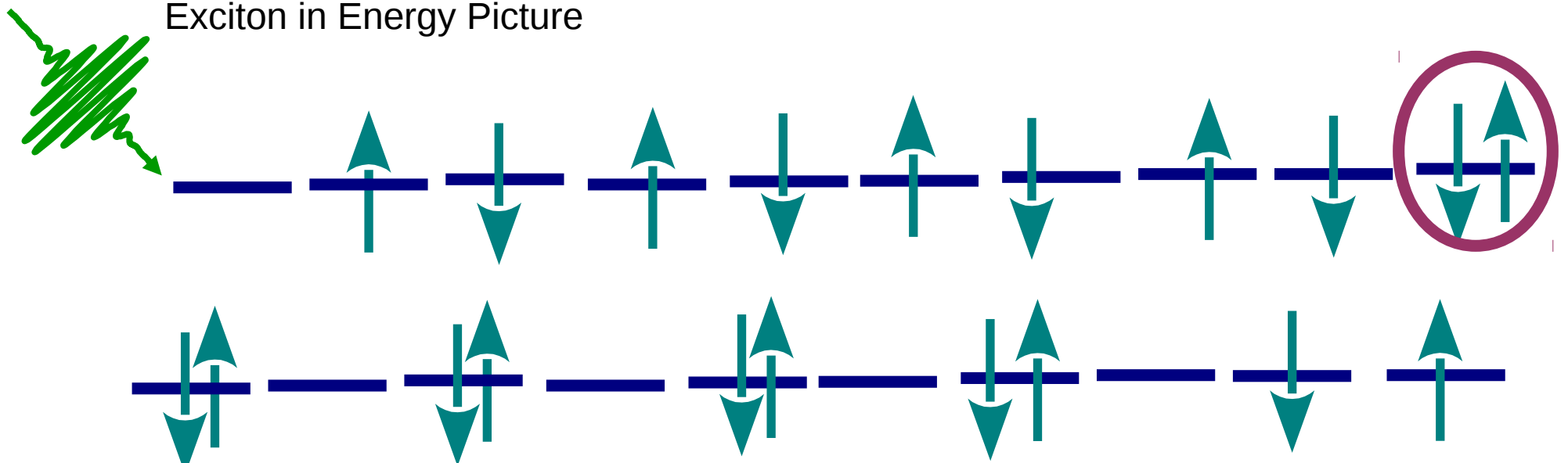
Exciton in Energy Picture



Exciton in Lattice Picture

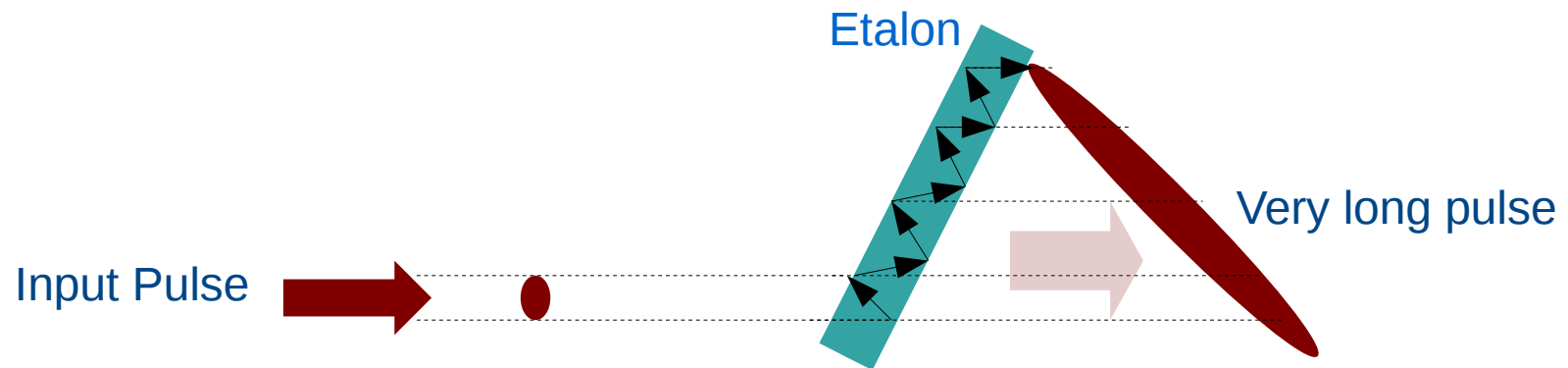


J. Phys. Chem. Lett., 2011, 2 (11), pp 1282–1288

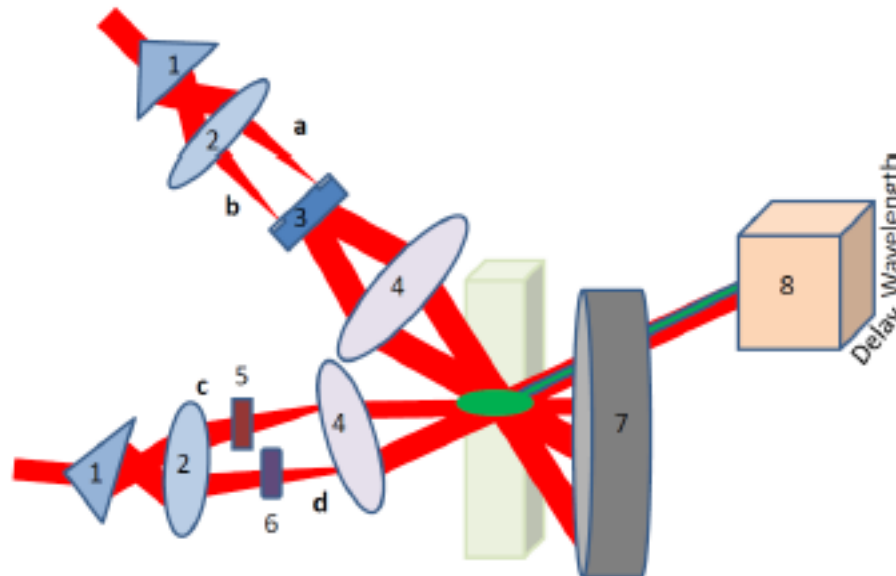


Future Outlook

We plan to build our own set up using the novel idea of pulse front tilt through etalon or bi-prism for single shot measurement to create long delays



Schematic for Single shot
Four Wave Mixing set up for
longer (ns) time scale
Measurement which has
not been explored yet!



Future Outlook

Setup & Instruments Required

1. Basic wet lab, fume hood with vacuum pump set up, inert gas cylinder, heating mantels, glass equipments etc. Facility for TEM, XRD.
2. Chemicals: Cadmium oxide, Te, Se powder, Zinc Oxide, oleic acid , 1octadecene, 1tetradecylphosphonic acid, Organic solvents (toluene, hexane, ethanol, acetone) etc, graphitic oxide,
3. For steady state study
Spectrophotometer, Fluorimeter, Cuvets etc.
4. For excited state and FRET study, TCSPC, Confocal Fluorescence Microscope
5. For 2D ultrafast optical coherent spectroscopy, we shall build our own set up. Pump laser, fs oscillator, chirp pulse amplifier, optical parametric oscillator, mirrors, spectrometer, CCD, prism, diffraction grating, etalon, holders, etc.
6. Computational facilities, Guassian, Matlab

Expertise

(Methods, Techniques and Tools):

Expertise (Methods, Techniques and Tools to be used):

a) Quantum dots synthesis, characterization by steady state spectroscopy, TEM, XRD.

b) Ultrafast spectroscopy, 2D electronic spectroscopy, Pump probe femto second laser set up,

(Coherent: MIRA900, LEGEND Elite, Optical parametric amplifier, TopasC light conversion, Pulse shaper Femto fit and MIIPS of Biophotonics, FROG, Swamp optics, Scanning Confocal

Fluorescence Microscope with pulsed picosecond lasers for fluorescence lifetime imaging.

Picoquant

Microtime 200)

c) Theoretical modeling, simulations, DFT calculations, many body code writing and calculations at CI level (Fortran, Gaussian, Matlab etc).

Publications in ultrafast spectroscopy and relevant fields

1. Tuning the Optically Bright and Dark States of Doped Graphene Quantum Dots
Mukhopadhyay, M.; Pandey, B.; Pati, S. K.
Phys. Rev. App. 2016, 6, 044014
2. Linear and Nonlinear Optical Properties of Graphene Quantum Dots:A Computational Study,
Yamijala S. SRKC; **Mukhopadhyay, M.**; Pati, S. K.
J. Phys. Chem. C, 2015, 119, 12079–12087
3. Coherent artifact study of twodimensional spectral shearing interferometry
Rhodes, M.; **Mukhopadhyay, M.**; Birge, J.; and Rick Trebino
Journal of the Optical Society of America B, 2015, 32 (9), 18811888
4. Rabi type oscillations in damped twodimensional single electron quantum dots,
Mukhopadhyay, M.; Hazra, R.K.; Ghosh, M.; Mukherjee, S.; Bhattacharyya, S.P
Cent. Eur. J. Phys, 2012, 10(4) , 983988.
5. Spectroscopic Signature of Intramolecular Charge Transfer Process in β Enaminones:A Combined Experimental and Theoretical Analysis,
Misra, R.; Mandal, A.; **Mukhopadhyay, M.**; Maity, D.K.; Bhattachayya, S.P
J. Phys. Chem. B, 2009, 113, 10779

Publications in ultrafast spectroscopy and relevant fields

6. Excited state intramolecular proton transfer in salicylidine –3,4,7methyl amine in some non polar and weakly polar solvents at room temperature and 77K and some studies on AM1 potential surfaces,

Mukhopadhyay, M.; Banerjee, D.; Koll, A.; Filarowski, A.; Guha, D.; Mukherjee, S. Indian J. Chem. A, 2006 , 45,1796.

7. Excited state intermolecular proton transfer and caging of alicylidine3,4,7methylamine in cyclodextrins,

Mukhopadhyay, M.; Banerjee, D.; Koll, A.; Mandal, A.; Filarowski, A.; Fitzmaurice, D.; Das,R.; Mukherjee, S. J. Photochem. Photobiol. A: Chem . , 2005 , 175 94.

8. Proton transfer reaction of a new orthohydroxy schiff base in some protic and aprotic solvents at room temperature and 77K,

Mukhopadhyay, M.; Banerjee, D.; Koll, A.; Filarowski, A.; Mukherjee, S. Spectrochim. Acta Part A , 2005 , 62 126.

9. Proton transfer reaction in 2hydroxy3formyl benzoic acid at room temperature and 77K and some study on AM1 potential surfaces,

Banerjee, D.; **Mukhopadhyay, M.**; Mukherjee, S. J. Photochem. Photobiol. A: Chem. 2005 , 172, 250.

Publications in ultrafast spectroscopy and relevant fields

10. 3D spectroscopy of coherences in colloidal quantum dots (in manuscript)

Mukhopadhyay, M.; Augulis, R.; Lenngren, N.; K.; Zidek, Zigmantas, D.; Pullerits, T.
(accepted for presentation in International Conference on Coherent Multidimensional Spectroscopy, 2012 http://www.mbi-berlin.de/cmds2012/images/program_cmds2012.pdf)

11. Tweaking optical efficiency by altering interfacial atom stoichiometry in hetero - structure nanocrystal,

Jose F. Aldana, **Madhuri Mukhopadhyay**, Benard Omogo, and Colin D. Heyes (in manuscript)

12. Nature of blinking with variation of shell materials on CdTe core quantum dot

Madhuri Mukhopadhyay, Feng Gao, Benard Omogo, and Colin D. Heyes (in manuscript)

13. The coherent artifact in modern pulse measurements. Rhodes, M.; **Mukhopadhyay, M.**; Birge, J.; Steinmeyer, G.; and RickTrebinio. Proc. SPIE, 2014, 8972, 89720R,

14. The coherent artifact in interferometric pulse-measurement techniques

Michelle Rhodes, **Madhuri Mukhopadhyay**, Jonathan Birge, Günter Steinmeyer, Saidur Rahaman, Rick Trebinio, Lasers and Electro-Optics (CLEO), 2014 Conference, published by IEEE

Publications in ultrafast spectroscopy and relevant fields

15. Nanopools governing proton transfer in diametrical ways in the ground and excited state, **Mukhopadhyay, M.**; Mandal, A.; Misra, R.; Banerjee, D.; Bhattacharyya, S.P .; Mukherjee, S. J. Phys. Chem. B, 2009 , 113, 567.

16. Effect of surfactant perturbed nanocaging on the ground and excited state proton transfer reaction, **Mukhopadhyay, M.**; Banerjee, D.; Mandal, A.; Bhattacharyya, S.P.; Mukherjee, S. J. Phys. Chem. A 2008 , 112, 12543.

17. Excited state proton transfer in di(2hydroxy3formyl5tertbutyl phenyl) methane and solvent effect, **Mukhopadhyay, M.**; Banerjee, D.; Mukherjee, S. Spectrochim. Acta Part A , 2006, 64, 287.

18. Ground and excited state proton transfer reaction of salicylidine3,4,7methyl amine in micelles, **Mukhopadhyay, M.**; Banerjee, D.; Koll, A.; Filarowski, A.; Mukherjee, S. Chem. Phys. Lett. 2006 , 420, 316.

19. Proton transfer reaction of 4methyl 2,6diformyl phenol in cyclodextrin nano cage, **Mukhopadhyay, M.**; Banerjee, D.; Mukherjee, S. J. Phys. Chem. A , 2006, 110, 12743.

Acknowledgement

IISC, Bangalore, Prof. P.K.Das

IACS,
Kolkata



Prof. S.P.
Bhattacharyya

Lund University,
Sweden



Prof. Tonu Pullerits

University of
Arkansas, USA



Prof. Colin D. Heyes

Georgia Institute
of Technology, USA



Prof. Rick Trebino

JNCASR,
Bangalore



Prof. Swapan K. Pati



Dr. Debi
Banerjee



Dr. Donatas
Zigmantas



Dr. Feng Gao



Dr. Michelle
Rhodes



Dr. Bradraj
Pandey



Dr. Sharma
S. R. K. C. Yamijala